

Statistical Regression

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TENTATIVE OUTLINE OF THE COURSE

- Simple Linear Regression
- Multiple Linear Regression
- Generalised Linear Models
- Nonlinear Regression
- Classification and Regression Trees
- *Random Forests*

BOOKS FOR PART 1

- Chatterjee, S. and Hadi, A.S. (2012). Regression Analysis by Example (Fifth Edition). Wiley.
- Dunn, P.K. and Smyth, G.K. (2018). Generalized Linear Models With Examples in R. Springer.
- Montgomery, D.C., Peck, E.A. and Vining G.G. (2012). Introduction to Linear Regression Analysis (Fifth Edition). Wiley.

WHAT IS REGRESSION ANALYSIS?

- Regression analysis: method for investigating functional relation among variables
- Relation expressed as an equation/model connecting the response (dependent) variable and one or more explanatory (independent) variables (or predictors)
 - Assessment by a real estate appraiser of sale price (response) of a house from its physical characteristics and paid taxes (explanatory)
 - Relation between cigarette consumption (response) and age, education, income, and their price (explanatory)
- Y response variable and $\underline{X} = (X_1, X_2, \dots, X_p)$, set of p explanatory variables
- True relation between Y and $\underline{X}$ approximated by regression model

$$Y = f(\underline{X}) + \varepsilon,$$

with ε random error representing the discrepancy in the approximation

- Linear regression: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \varepsilon$

STEPS IN REGRESSION ANALYSIS

- Statement of the problem
- Selection of potentially relevant variables
- Data collection
- Model specification
- Choice of fitting method
- Model fitting
- Model validation and criticism
- Using the chosen model(s) for the solution of the posed problem

STEPS IN REGRESSION ANALYSIS

- Statement of the problem
 - Identifying *right* questions to be addressed avoids choice of *wrong* models and *wrong* answers
 - If interested in employment discrimination about gender, there is a difference between the questions
 - * On the average, are women paid less than equally qualified men?
 - * On the average, are women more qualified than equally paid men?
 - In the first case, salary is the response and qualification and gender are explanatory variables, while in the second qualification is the response and the others are explanatory variables
- Selection of potentially relevant variables
 - Response variable: price of a single family house in a given geographical area
 - Explanatory variables: area of the lot, area of the house, age of the house, number of bedrooms, number of bathrooms, type of neighbourhood, style of the house, amount of real estate taxes, etc.

STEPS IN REGRESSION ANALYSIS

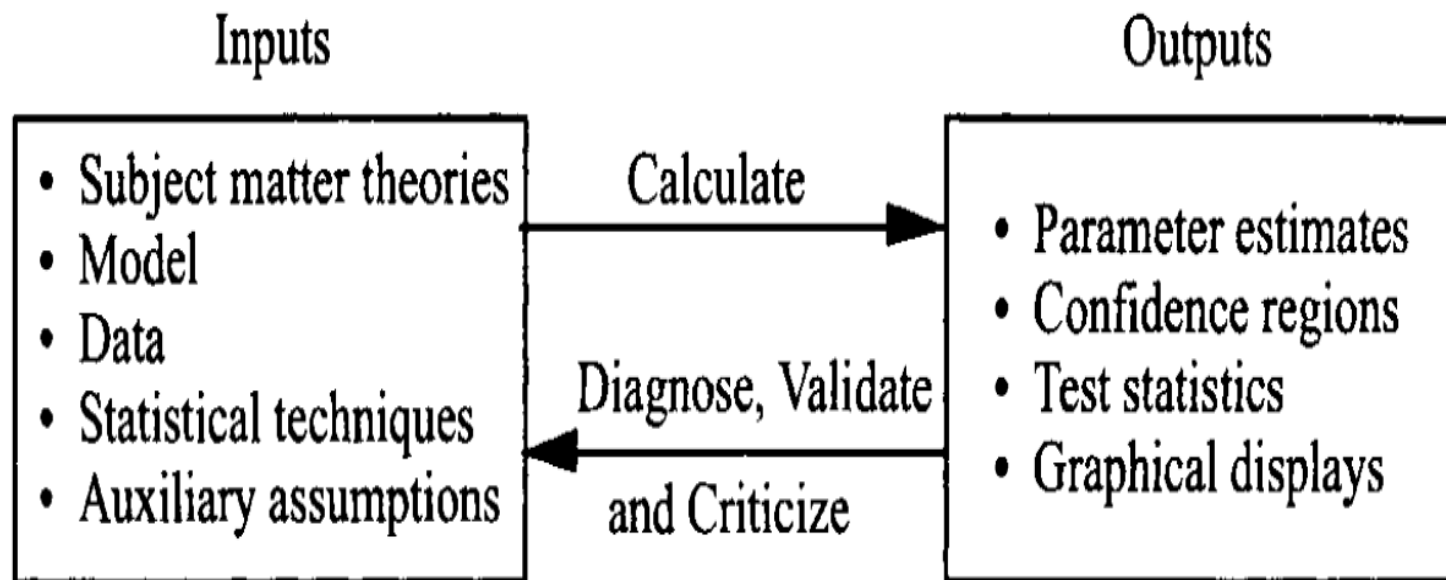
- Data collection
 - Observational data: not controlled by experimenter, e.g. hourly temperatures
 - Experimental data, e.g. randomised clinical trials
 - Missing and censored data, outliers
 - Qualitative (e.g. bad/good neighbourhood) and quantitative (e.g. price of the house) data
- Model specification
 - Choice of the functional form (e.g. linear/nonlinear) of the relation between response and explanatory variables
 - * $Y = \beta_0 + \beta_1 X_1 + \varepsilon$ vs. $Y = \beta_0 + e^{\beta_1 X_1} + \varepsilon$
 - Choice of the parameters: which and how many?
 - Response variable: one or more? E.g.: height and weight as function of age and gender or weight as function of height, age and gender

STEPS IN REGRESSION ANALYSIS

- Choice of fitting method
 - (Weighted) least square methods
 - Statistical methods
 - * Maximum likelihood estimation
 - * Bayesian estimation
- Model fitting
 - Parameters replaced by their estimates
 - Predicted value of response variable for a new set of explanatory variables
 - In general, better not to predict response variables for values of the explanatory variables far outside the range of actual data

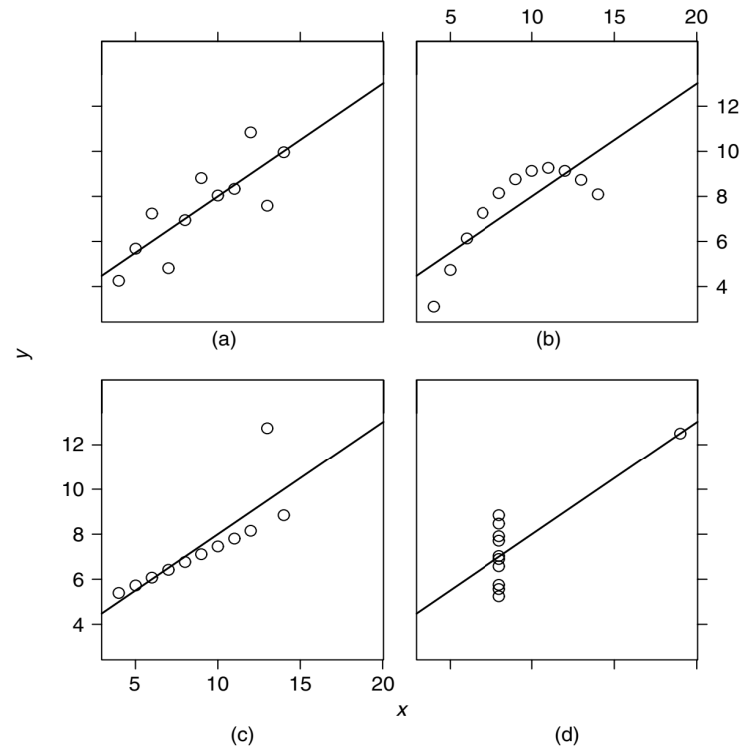
STEPS IN REGRESSION ANALYSIS

- Model validation and criticism
 - The choice of a model has to be justified a priori based on the knowledge and the available data, but it has also to be validated a posteriori showing that it fits the data well and it is suitable to pursue the stated goals
 - It is an iterative process, like in the figure below from Chatterjee and Hadi (2006)



STEPS IN REGRESSION ANALYSIS

- Model validation and criticism
 - Same estimated model but different data: when is it a good model? Example due to Anscombe (1973)

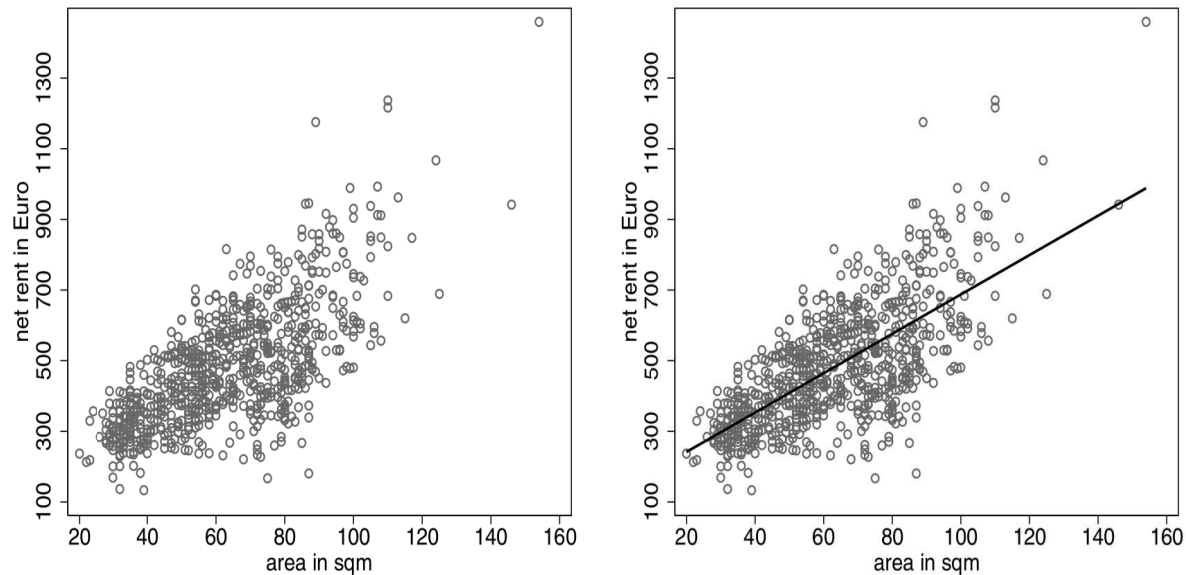


STEPS IN REGRESSION ANALYSIS

- Using the chosen model(s) for the solution of the posed problem
 - Determine if an explanatory variable (e.g. number of bedrooms) has an effect on the response variable (value of a house) and in which direction (increase or decrease of the value)
 - Analyse the effects of policy that involves changing values of the explanatory variables (e.g. probability of repayment of a mortgage if its duration goes from 20 to 30 years)
 - Forecast values of the response variable for a given set of new explanatory values (probability of repayment of the mortgage by a new customer)

STEPS IN REGRESSION ANALYSIS

- From scatterplot (left) to a model (right)
- Relation between rent (response) and area in square meters (explanatory) of apartments in Munich, Germany, from Fahrmeir et al. (2022)



CORRELATION

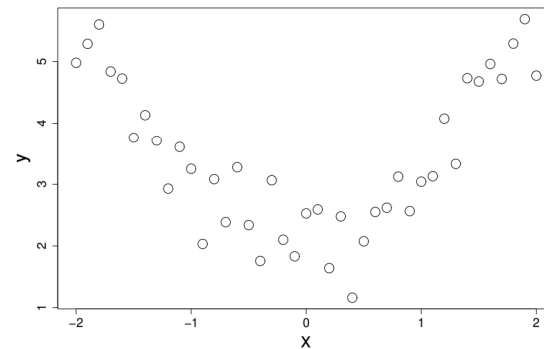
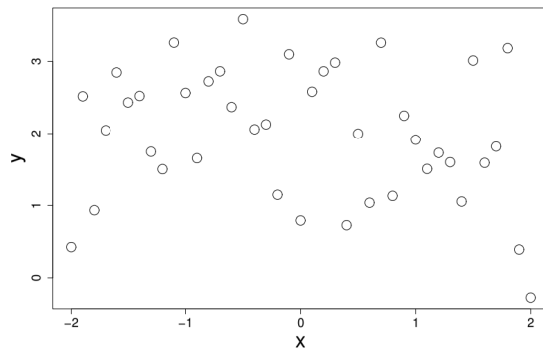
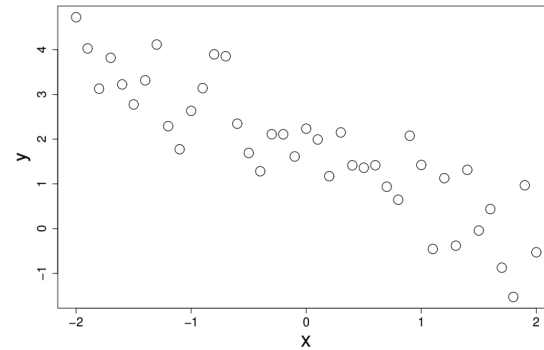
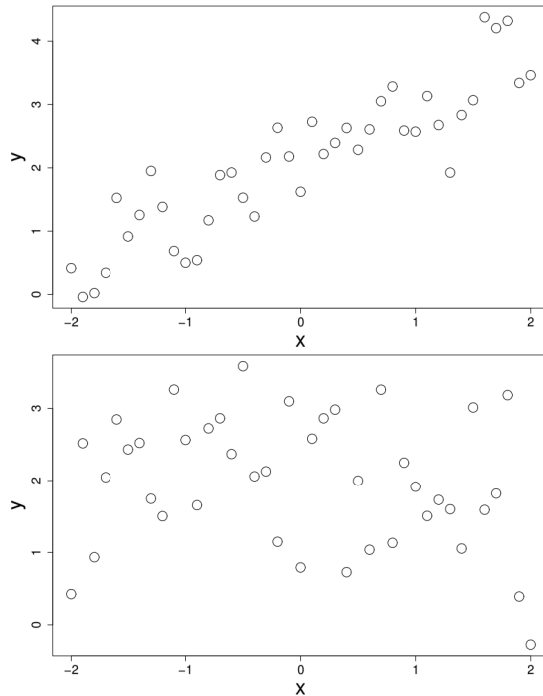
- X and Y r.v.'s with (discrete/continuous) densities f_X and f_Y
- g function s.t.
 - $E(g(X)) = \int g(x)f_X(x)dx$ (X continuous) or $= \sum_x g(x)f_X(x)$ (X discrete)
 - $Var(g(X)) = E[(g(X) - E(g(X)))^2]$
- Properties
 - $E(aX + b) = aE(X) + b$ for $a, b \in \mathbb{R}$
 - $Var(aX + b) = a^2Var(X)$ for $a, b \in \mathbb{R}$
 - $Cov(X, Y) = E[(X - E(X))(Y - E(Y))]$
 - $E(X + Y) = E(X) + E(Y)$
 - $Var(X \pm Y) = Var(X) + Var(Y) \pm 2Cov(X, Y)$
 - X and Y independent $\Rightarrow E(XY) = E(X)E(Y)$
 - $Var(X) = E(X^2) - [E(X)]^2$

CORRELATION

- Correlation between X and Y : $\rho = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$
 - Correlation between -1 and 1
 - * $+1$: Perfect positive (direct) linear relation
 - * -1 : Perfect negative (inverse) linear relation
 - * 0 : No linear relation
- Sample correlation for paired data $(x_1, y_1), \dots, (x_n, y_n)$ from
 - Sample means $\bar{x} = \sum_{i=1}^n \frac{x_i}{n}$ and $\bar{y} = \sum_{i=1}^n \frac{y_i}{n}$
 - Sample variances $Var(X) = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n - 1}$ and $Var(Y) = \sum_{i=1}^n \frac{(y_i - \bar{y})^2}{n - 1}$
 - Sample covariance $Cov(X, Y) = \sum_{i=1}^n \frac{(x_i - \bar{x})(y_i - \bar{y})}{n - 1}$

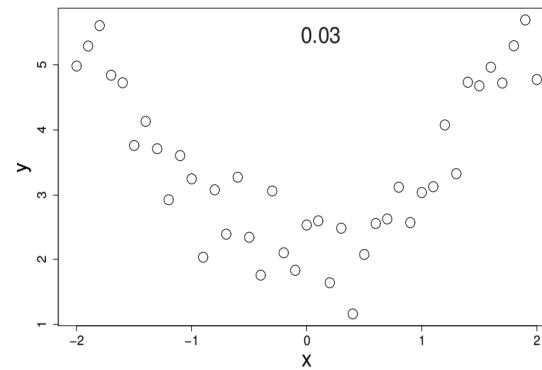
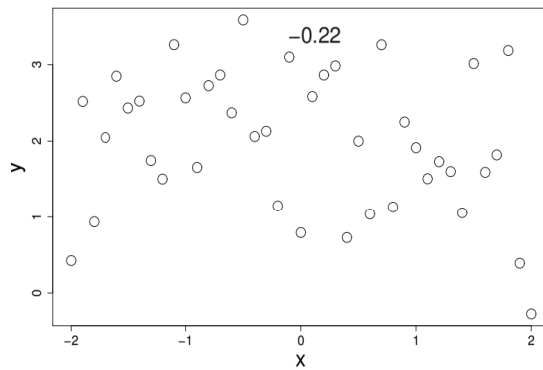
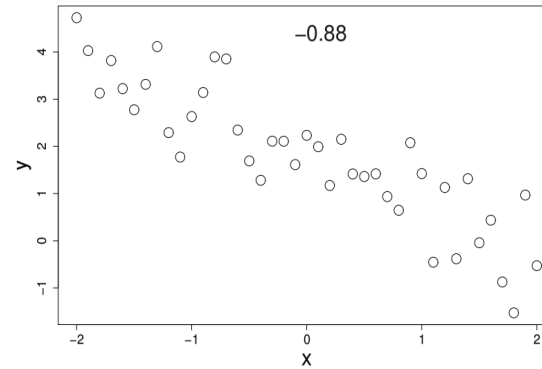
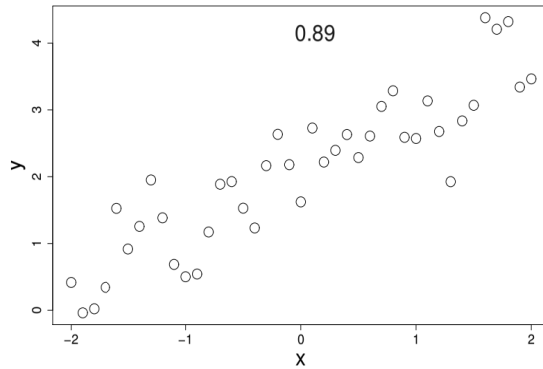
CORRELATION

- What can we tell about the correlation between the variables in those examples (due to Biatat, AIMS 2024)?



CORRELATION

- What can we tell about the correlation between the variables in those examples (due to Biatat, AIMS 2024)?



EXERCISE 1

- Consider the dataset `mtcars` (Motor Trend Car Road Tests) in R
- Data from 1974 Motor Trend US magazine, about fuel consumption and 10 aspects of automobile design and performance for 32 automobiles

```
mtcars
```

```
head(mtcars)
```

```
attach(mtcars) #Object with column name (mpg instead of mtcars$mpg)
```

<code>mpg</code>	Miles/(US) gallon
<code>cyl</code>	Number of cylinders
<code>disp</code>	Displacement (cubic inches)
<code>hp</code>	Gross horsepower
<code>drat</code>	Rear axle ratio
<code>wt</code>	Weight (1000 lbs)
<code>qsec</code>	Second to ride 1/4 mile time
<code>vs</code>	Engine (0 = V-shaped, 1 = straight)
<code>am</code>	Transmission (0 = automatic, 1 = manual)
<code>gear</code>	Number of forward gears
<code>carb</code>	Number of carburetors

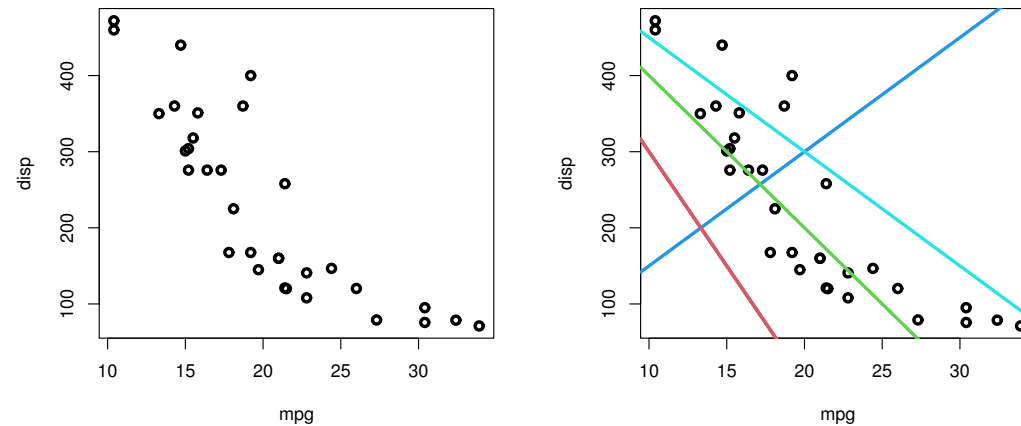
EXERCISE 1

- Consider `mpg` as response and plot it against all the other (explanatory) variables, one at the time and comment on possible correlation
- Compute the correlation between `mpg` and all the explanatory variables using the R command and comment
- Use the previous formulas to compute the correlation between `mpg` and `disp` and compare with the previous result
- Use the formulas to compute mean, sample variance and sample covariance for `mpg` and `disp`, compare with the R commands and comment

CORRELATION AND REGRESSION

- Limitations
 - Correlation coefficient quantifies magnitude of linear associations
 - What about non-linear associations?
 - What about categorical variables?
 - What about 3+ variables?
 - What about predictions?
 - ...
- Regression allows to overcome these limitations and to do more

SIMPLE LINEAR REGRESSION



- How to model the relation between miles per gallon (mpg) and displacement ($disp$)?
- We try a linear relation $\Rightarrow$ (simple) linear regression
- How to choose the parameters? We will see three methods
 - Least squares
 - Maximum likelihood
 - Bayesian

REGRESSION

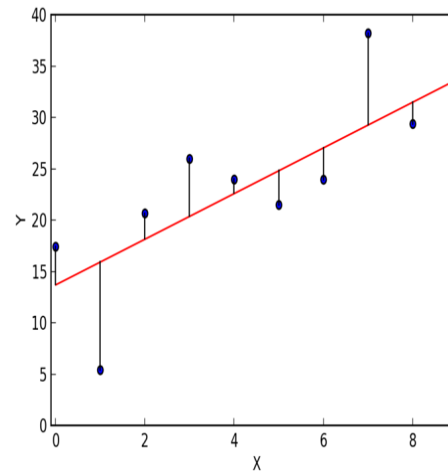
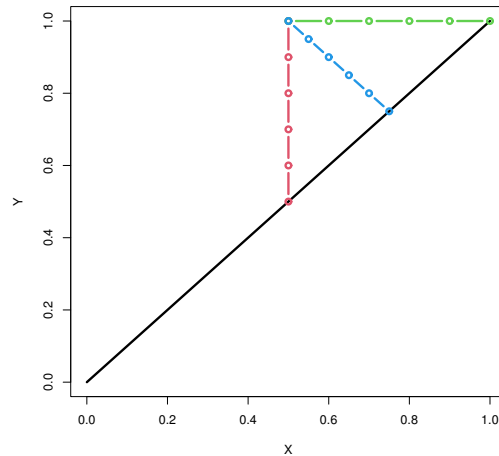
- We now consider linear regression (LR), providing a linear relation between a dependent variable (Y) and an independent one (X), sometimes called *covariate*
- We can distinguish 4 cases based on the dimensions of Y and X
 - Simple LR vs. Multiple LR: just one X or multiple X 's
 - Univariate LR vs. Multivariate LR: just one-dimensional Y or multiple dimensional Y
- We consider now the simplest case: Univariate Simple Linear Regression
- $Y = \beta_0 + \beta_1 X + \varepsilon$
- β_0, β_1 univariate unknown parameters
- ε error term with $E(\varepsilon) = 0$ and $Var(\varepsilon) = \sigma^2$ unknown
- Later we consider $\varepsilon \sim \mathcal{N}(0, \sigma^2)$

SIMPLE LINEAR REGRESSION

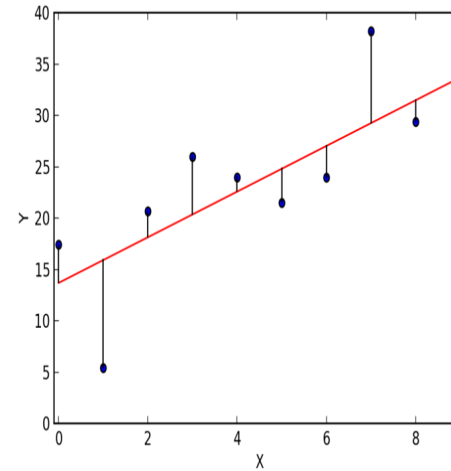
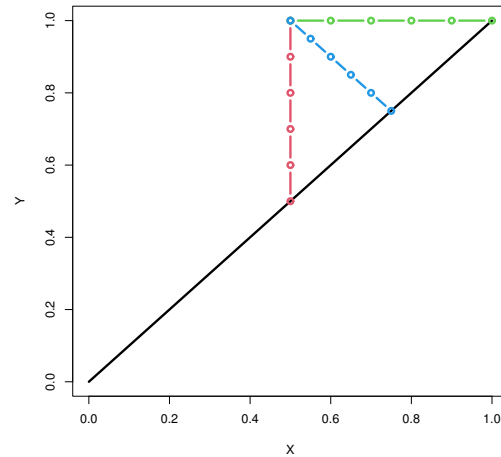
- Simple linear regression model: $Y = \beta_0 + \beta_1 X + \varepsilon$
 - Regression coefficients: β_0 (intercept) and β_1 (slope)
 - ε : random error component with $E(\varepsilon) = 0$ and $Var(\varepsilon) = \sigma^2$ unknown
 - Uncorrelated ε for different observations (y, x)
- What are the interpretation of β_0 and β_1 ?
- Is $\beta_1 = 1$ small? What about $\beta_1 = 1,000,000$?
- Y r.v. while X exactly known, with no measurement error/variability
- $E(Y|X) = E(\beta_0 + \beta_1 X + \varepsilon|X) = \beta_0 + \beta_1 X$
- $Var(Y|X) = Var(\beta_0 + \beta_1 X + \varepsilon|X) = Var(\varepsilon) = \sigma^2$
- Uncorrelated $\varepsilon \Rightarrow$ uncorrelated Y

SIMPLE LINEAR REGRESSION

- Our goals: estimate β_0 , β_1 , σ^2 and forecast Y for new X
- We start making no distributional assumption about ε
- We look for the "best" linear function expressing the relation between X and Y
- What do we mean by "best"?



SIMPLE LINEAR REGRESSION



- We can see (left) many ways to measure the distance between the point (Y) and the line ($\beta_0 + \beta_1 X$)
- Which one to choose?
- We are going to see soon that we choose the vertical distance (right)
- How are we considering the vertical distances? With sign or squared?

SIMPLE LINEAR REGRESSION

- Data: $(y_1, x_1), \dots, (y_n, x_n)$
- Data could come from historical records or from a controlled experiment planned to collect them, but this is not relevant for us now
- We now present the **Least Squares Method** to estimate the parameters β_0 and β_1 when no distributional assumption is made about ε but $E(\varepsilon) = 0$ and $Var(\varepsilon) = \sigma^2$
- We estimate β_0 and β_1 so that the sum of the squares of the differences between the observations y_i and the straight line is a minimum
- Population regression model: $Y = \beta_0 + \beta_1 X + \varepsilon$
- Sample regression model: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, i = 1, \dots, n$

SIMPLE LINEAR REGRESSION

- Least square method $\Rightarrow$ Minimise $S(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$
- $\hat{\beta}_0$ and $\hat{\beta}_1$ least squares estimators (LSEs) of β_0 and β_1 s.t.

$$\left. \frac{\partial S}{\partial \beta_0} \right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$
$$\left. \frac{\partial S}{\partial \beta_1} \right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) x_i = 0$$

- Simplifying, we get the least squares normal equations:

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n x_i = \sum_{i=1}^n y_i$$
$$\hat{\beta}_0 \sum_{i=1}^n x_i + \hat{\beta}_1 \sum_{i=1}^n x_i^2 = \sum_{i=1}^n y_i x_i$$

SIMPLE LINEAR REGRESSION

- Least squares normal equations:

$$\begin{aligned}n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n x_i &= \sum_{i=1}^n y_i \\ \hat{\beta}_0 \sum_{i=1}^n x_i + \hat{\beta}_1 \sum_{i=1}^n x_i^2 &= \sum_{i=1}^n y_i x_i\end{aligned}$$

- $\hat{\beta}_0$ and $\hat{\beta}_1$ LSE given by

$$\begin{aligned}\hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} \\ \hat{\beta}_1 &= \left\{ \sum_{i=1}^n y_i x_i - \frac{\left(\sum_{i=1}^n y_i \right) \left(\sum_{i=1}^n x_i \right)}{n} \right\} / \left\{ \sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i \right)^2}{n} \right\}\end{aligned}$$

$$\text{where } \bar{y} = \frac{\sum_{i=1}^n y_i}{n} \text{ and } \bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

SIMPLE LINEAR REGRESSION

- $$\hat{\beta}_1 = \left\{ \sum_{i=1}^n y_i x_i - \frac{\left(\sum_{i=1}^n y_i \right) \left(\sum_{i=1}^n x_i \right)}{n} \right\} / \left\{ \sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i \right)^2}{n} \right\}$$

- $\Rightarrow$ simplified as $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$ with

$$S_{xx} = \sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i \right)^2}{n} = \sum_{i=1}^n (x_i - \bar{x})^2$$
$$S_{xy} = \sum_{i=1}^n y_i x_i - \frac{\left(\sum_{i=1}^n y_i \right) \left(\sum_{i=1}^n x_i \right)}{n} = \sum_{i=1}^n y_i (x_i - \bar{x})$$

SIMPLE LINEAR REGRESSION

- $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$ Corrected sum of squares of x_i 's
- $S_{xy} = \sum_{i=1}^n y_i(x_i - \bar{x})$ Corrected sum of cross products of x_i 's and y_i 's
- Fitted simple regression model: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$
- We can plug-in a new value x or an observed x_i
- In the former case we make a prediction while in the latter we compute the difference between the observed y_i and the fitted $\hat{y}_i$
- $\Rightarrow i$ -th residual $e_i = y_i - \hat{y}_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i), i = 1, \dots, n$
- Residuals are important to investigate model adequacy and detect departure from underlying assumptions (we will see later)

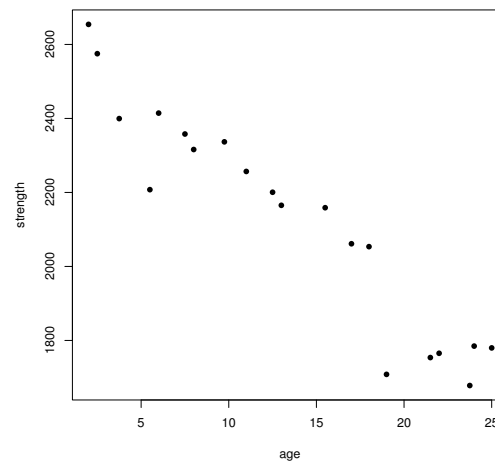
ROCKET PROPELLANT DATA*

A rocket motor is manufactured by bonding an igniter propellant and a sustainer propellant together inside a metal housing. The shear strength of the bond between the two types of propellant is an important quality characteristic. It is suspected that shear strength is related to the age in weeks of the batch of sustainer propellant. Twenty observations on shear strength and the age of the corresponding batch of propellant have been collected and are shown in the following table and available in the file `propellant.txt`

strength	2158.70	1678.15	2316.00	2061.30	2207.50	1708.30	1784.70
age	15.50	23.75	8.00	17.00	5.50	19.00	24.00
strength	2575.00	2357.90	2256.70	2165.20	2399.55	1779.80	2336.75
age	2.50	7.50	11.00	13.00	3.75	25.00	9.75
strength	1765.30	2053.50	2414.40	2200.50	2654.20	1753.70	
age	22.00	18.00	6.00	12.50	2.00	21.50	

**From Montgomery, Peck and Vining, Introduction to Linear Regression Analysis, 5th Ed., Wiley*

ROCKET PROPELLANT DATA



- The plot suggests a strong relation between shear strength and propellant age
- $\Rightarrow$ We try a linear model!

ROCKET PROPELLANT DATA

$$S_{xx} = \sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n} = 4677.69 - \frac{71422.56}{20} = 1106.56$$

$$S_{xy} = \sum_{i=1}^n y_i x_i - \frac{\left(\sum_{i=1}^n y_i\right)\left(\sum_{i=1}^n x_i\right)}{n} = 528492.64 - \frac{267.25 \cdot 42627.15}{20} = -41112.65$$

Least squares estimators:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{-41112.65}{1106.56} = -37.15$$

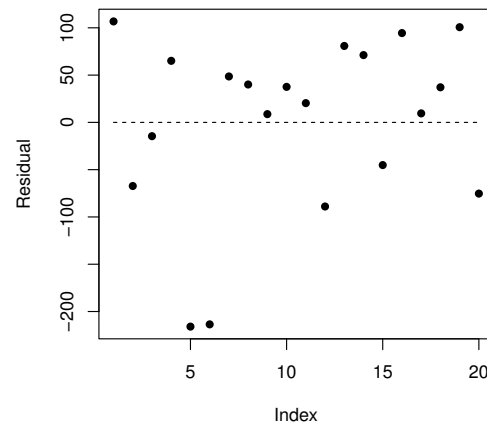
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 2131.3575 - (-37.15) \cdot 133625 = 2627.82$$

- $\Rightarrow$ Least squares fit: $\hat{y} = 2627.82 - 37.15x$
- Slope -37.15 : average weekly decrease in shear strength due to propellant age
- Intercept 2627.82 : shear strength in a batch of propellant right after manufacture

ROCKET PROPELLANT DATA

Observed Value, y_i	Fitted Value, $\hat{y}_i$	Residual, e_i
2158.70	2051.94	106.76
1678.15	1745.42	-67.27
2316.00	2330.59	-14.59
2061.30	1996.21	65.09
2207.50	2423.48	-215.98
1708.30	1921.90	-213.60
1784.70	1736.14	48.56
2575.00	2534.94	40.06
2357.90	2349.17	8.73
2256.70	2219.13	37.57
2165.20	2144.83	20.37
2399.55	2488.50	-88.95
1799.80	1698.98	80.82
2336.75	2265.58	71.17
1765.30	1810.44	-45.14
2053.50	1959.06	94.44
2414.40	2404.90	9.50
2200.50	2163.40	37.10
2654.20	2553.52	100.68
1753.70	1829.02	-75.32
$\Sigma y_i = 42,627.15$	$\Sigma \hat{y}_i = 42,627.15$	$\Sigma e_i = 0.00$

ROCKET PROPELLANT DATA



- How well does this equation fit the data?
- Are the assumptions about constant variance and uncorrelated errors meaningful?
- We should expect a cloud around 0 but here it does not seem the case

SIMPLE LINEAR REGRESSION

- $\hat{\beta}_0$ and $\hat{\beta}_1$ linear combination of observations y_i

$$- \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \sum_{i=1}^n c_i y_i \text{ with } c_i = (x_i - \bar{x})/S_{xx}$$

$$- \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{1}{n} \sum_{i=1}^n y_i - \bar{x} \sum_{i=1}^n c_i y_i = \sum_{i=1}^n \left(\frac{1}{n} - \bar{x} c_i \right) y_i$$

- $\hat{\beta}_1$ unbiased estimator

$$\begin{aligned} E(\hat{\beta}_1) &= E\left(\sum_{i=1}^n c_i y_i\right) = \sum_{i=1}^n c_i E(y_i) = \sum_{i=1}^n c_i (\beta_0 + \beta_1 x_i) = \\ &= \beta_0 \sum_{i=1}^n c_i + \beta_1 \sum_{i=1}^n c_i x_i = \beta_1 \end{aligned}$$

$$\text{since } \sum_{i=1}^n c_i = \sum_{i=1}^n \frac{(x_i - \bar{x})}{S_{xx}} = 0 \text{ and } \sum_{i=1}^n c_i x_i = \frac{\sum_{i=1}^n (x_i - \bar{x}) x_i}{S_{xx}} = 1$$

SIMPLE LINEAR REGRESSION

- $\hat{\beta}_0$ unbiased estimator

$$E(\hat{\beta}_0) = E(\bar{y} - \hat{\beta}_1 \bar{x}) = E\left(\sum_{i=1}^n \frac{y_i}{n}\right) - \beta_1 \bar{x} = \sum_{i=1}^n \frac{\beta_0 + \beta_1 x_i}{n} - \beta_1 \bar{x} = \beta_0$$

- Variance of $\hat{\beta}_1$ (using uncorrelated y_i and $Var(y_i) = \sigma^2$)

$$Var(\hat{\beta}_1) = Var\left(\sum_{i=1}^n c_i y_i\right) = \sum_{i=1}^n c_i^2 Var(y_i) = \sigma^2 \sum_{i=1}^n c_i^2 = \frac{\sigma^2 \sum_{i=1}^n (x_i - \bar{x})^2}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}}$$

- Variance of $\hat{\beta}_0$ (using $Cov(\bar{y}, \hat{\beta}_1) = 0$)

$$Var(\hat{\beta}_0) = Var(\bar{y} - \hat{\beta}_1 \bar{x}) = Var(\bar{y}) + \bar{x}^2 Var(\hat{\beta}_1) - 2\bar{x} Cov(\bar{y}, \hat{\beta}_1) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)$$

SIMPLE LINEAR REGRESSION

- The estimate of σ^2 is obtained from the error (or residual) sum of squares:

$$SS_{Res} = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- Using $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ and replacing $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i = (\bar{y} - \hat{\beta}_1 \bar{x}) + \hat{\beta}_1 x_i$, we get

$$\begin{aligned} SS_{Res} &= \sum_{i=1}^n [(y_i - \bar{y}) - \hat{\beta}_1(x_i - \bar{x})]^2 \\ &= \sum_{i=1}^n (y_i - \bar{y})^2 - 2\hat{\beta}_1 \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) + \hat{\beta}_1^2 \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= SS_T - 2\hat{\beta}_1 \sum_{i=1}^n y_i(x_i - \bar{x}) + 2\hat{\beta}_1 \bar{y} \sum_{i=1}^n (x_i - \bar{x}) + \hat{\beta}_1^2 \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= SS_T - 2\hat{\beta}_1 S_{xy} + 0 + \hat{\beta}_1^2 S_{xx} = SS_T - \hat{\beta}_1 S_{xy} = SS_T - \frac{S_{xy}^2}{S_{xx}} \end{aligned}$$

- with $SS_T = \sum_{i=1}^n (y_i - \bar{y})^2$ and $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$

SIMPLE LINEAR REGRESSION

- It is possible to prove that $E(SS_{Res}) = (n - 2)\sigma^2 \Rightarrow \hat{\sigma}^2 = \frac{SS_{Res}}{n - 2} = MS_{Res}$ unbiased estimator of σ^2
- MS_{Res} residual mean square
- $\sqrt{\hat{\sigma}^2}$ standard error of regression, in the same unit as response y
- $n - 2$ is the number of degrees of freedom for the residual sum of squares SS_{Res} since two degrees of freedom are associated with the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$
- Since $\hat{\sigma}^2$ depends on the residual sum of squares, any violation of the assumptions on the model errors or any misspecification of the model form may seriously damage the usefulness of $\hat{\sigma}^2$ as an estimate of σ^2
- Since $\hat{\sigma}^2$ is computed from the regression model residuals, we say that it is a model-dependent estimate of σ^2

SIMPLE LINEAR REGRESSION

- Interest in testing hypotheses and constructing confidence intervals
- Need to assume independent errors $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$
- Test: $H_0 : \beta_1 = \beta_{10}$ vs. $H_1 : \beta_1 \neq \beta_{10}$
- $\varepsilon \sim \mathcal{N}(0, \sigma^2) \Rightarrow y_i \sim \mathcal{N}(\beta_0 + \beta_1 x_i, \sigma^2)$
- $\hat{\beta}_1 = \sum_{i=1}^n c_i y_i$ with $c_i = (x_i - \bar{x})/S_{xx} \Rightarrow \hat{\beta}_1 \sim \mathcal{N}(\beta_1, \sigma^2/S_{xx})$
- Under the null hypothesis $H_0 \Rightarrow Z_0 = \frac{\hat{\beta}_1 - \beta_{10}}{\sqrt{\sigma^2/S_{xx}}} \sim \mathcal{N}(0, 1)$
- Since σ^2 is unknown $\Rightarrow t_0 = \frac{\hat{\beta}_1 - \beta_{10}}{\sqrt{MS_{Res}/S_{xx}}} \sim t_{n-2}$
- Reject H_0 if $|t_0| > t_{1-\alpha/2, n-2}$, quantile of order $1 - \alpha/2$

SIMPLE LINEAR REGRESSION

- Similarly, $H_0 : \beta_0 = \beta_{00}$ vs. $H_1 : \beta_0 \neq \beta_{00}$

- $\Rightarrow t_0 = \frac{\hat{\beta}_0 - \beta_{00}}{\sqrt{MS_{Res}(1/n + \bar{x}^2/S_{xx})}} \sim t_{n-2}$

- Important test about (lack of) linear relation between y and x
 $H_0 : \beta_1 = 0$ vs. $H_1 : \beta_1 \neq 0$

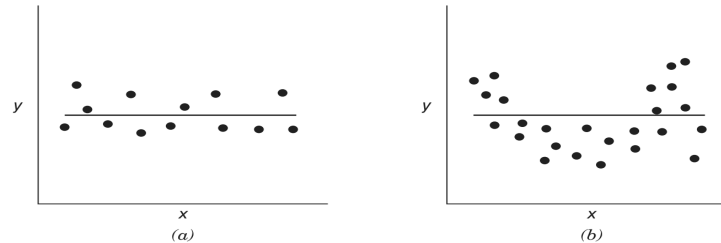


Figure 2.2 Situations where the hypothesis $H_0: \beta_1 = 0$ is not rejected.

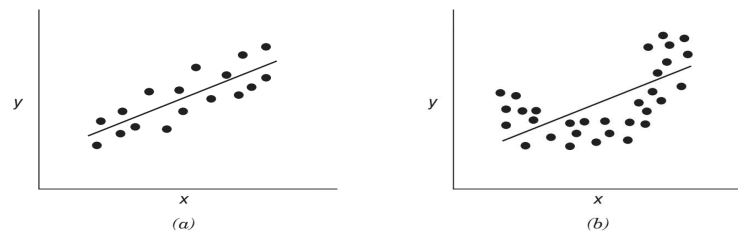


Figure 2.3 Situations where the hypothesis $H_0: \beta_1 = 0$ is rejected.

SIMPLE LINEAR REGRESSION

- 100(1 - α)% Confidence intervals

$$- \hat{\beta}_1 - t_{1-\alpha/2, n-2} \sqrt{MS_{Res}/S_{xx}} \leq \beta_1 \leq \hat{\beta}_1 + t_{1-\alpha/2, n-2} \sqrt{MS_{Res}/S_{xx}}$$

$$- \hat{\beta}_0 - t_{1-\alpha/2, n-2} \sqrt{MS_{Res}(1/n + \bar{x}^2/S_{xx})} \leq \beta_0 \leq \hat{\beta}_0 + t_{1-\alpha/2, n-2} \sqrt{MS_{Res}(1/n + \bar{x}^2/S_{xx})}$$

$$- \frac{(n-2)MS_{Res}}{\chi_{1-\alpha/2, n-2}^2} \leq \sigma^2 \leq \frac{(n-2)MS_{Res}}{\chi_{\alpha/2, n-2}^2}$$

- Prediction of new observation at x_0 : $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$

- 100(1 - α)% Prediction interval

$$\hat{y}_0 - t_{1-\alpha/2, n-2} \sqrt{MS_{Res} \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)} \leq y_0 \leq \hat{y}_0 + t_{1-\alpha/2, n-2} \sqrt{MS_{Res} \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}$$

SIMPLE LINEAR REGRESSION

- Coefficient of determination $R^2 = 1 - \frac{SS_{Res}}{SS_T}$
- Remember: $SS_{Res} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$ and $SS_T = \sum_{i=1}^n (y_i - \bar{y})^2$
- SS_T measures variability in y without considering the effect of x , while SS_{Res} measures variability remaining in y after considering x
- $0 \leq SS_{Res} \leq SS_T \Rightarrow 0 \leq R^2 \leq 1$
- Values of R^2 close to 1 $\Rightarrow$ most variability in y explained by regression model
- The statistic R^2 could be used with caution
 - A polynomial of degree $n - 1$ gives $R^2 = 1$ but it is not good for predictions
 - R^2 might be large but the relation is not linear

ROCKET PROPELLANT DATA

- $SS_T = 1693737.60$
- $SS_{Res} = SS_T - \hat{\beta}_1 S_{xy} = 166402.65$
- $\Rightarrow \hat{\sigma}^2 = \frac{SS_{Res}}{n - 2} = \frac{166402.65}{18} = 9244.59$
- $H_0 : \beta_1 = 0$ vs. $H_1 : \beta_1 \neq 0$
 - $t_0 = \frac{\hat{\beta}_1}{\sqrt{MS_{Res}/S_{xx}}} = \frac{-37.15}{2.89} = -12.85$
 - $t_{.975,18} = 2.101 \Rightarrow \text{reject } H_0$
 - $\Rightarrow$ there is linear relation between shear strength and propellant age
- 95% Confidence interval for $\beta_1 : (-43.22, -31.08)$
- Shear strength in motor with 10 weeks old propellant: $2048.32 \leq y_0 \leq 2464.32$
- $R^2 = 0.9018$

SIMPLE LINEAR REGRESSION

Some critical aspects

- Regression models are intended as interpolation equations over the range of x while extrapolation should be done very carefully
- Slope more influenced by remote values of x
- Outliers can disturb the least squares fit
- Correlation does not imply causation
- Covariates might not be known exactly, e.g., as a result of another prediction process or measured with some imprecision

SIMPLE LINEAR REGRESSION

- Observations: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, i = 1, \dots, n$
- x_i 's are supposed known here but they could be r.v.'s as well
- We assume that $\varepsilon_1, \dots, \varepsilon_n$ are i.i.d. $\mathcal{N}(0, \sigma^2)$
- Notation: $\underline{y} = (y_1, \dots, y_n)$ and $\underline{x} = (x_1, \dots, x_n)$
- Likelihood function $L(\beta_0, \beta_1, \sigma^2 | \underline{y}, \underline{x})$ given by

$$\begin{aligned} \prod_{i=1}^n f(y_i | x_i, \beta_0, \beta_1, \sigma^2) &= \prod_{i=1}^n \left\{ \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right\} \right\} \\ &= \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left\{-\frac{\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right\} \end{aligned}$$

- Look for values of parameters maximising the likelihood
 $\Rightarrow$ Maximum likelihood estimators (MLEs) $\tilde{\beta}_0, \tilde{\beta}_1$ and $\tilde{\sigma}^2$

SIMPLE LINEAR REGRESSION

- Consider log-likelihood $l(\beta_0, \beta_1, \sigma^2 | \underline{y}, \underline{x}) = \log L(\beta_0, \beta_1, \sigma^2 | \underline{y}, \underline{x})$
 $\Rightarrow$ same MLEs but simpler computations

- $$l(\beta_0, \beta_1, \sigma^2 | \underline{y}, \underline{x}) = -\frac{n}{2} \log 2\pi - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

- MLEs $\tilde{\beta}_0, \tilde{\beta}_1$ and $\tilde{\sigma}^2$ should satisfy

$$\left. \frac{\partial l}{\partial \beta_0} \right|_{\tilde{\beta}_0, \tilde{\beta}_1, \tilde{\sigma}^2} = \frac{1}{\tilde{\sigma}^2} \sum_{i=1}^n (y_i - \tilde{\beta}_0 - \tilde{\beta}_1 x_i) = 0$$

$$\left. \frac{\partial l}{\partial \beta_1} \right|_{\tilde{\beta}_0, \tilde{\beta}_1, \tilde{\sigma}^2} = \frac{1}{\tilde{\sigma}^2} \sum_{i=1}^n (y_i - \tilde{\beta}_0 - \tilde{\beta}_1 x_i) x_i = 0$$

$$\left. \frac{\partial l}{\partial \sigma^2} \right|_{\tilde{\beta}_0, \tilde{\beta}_1, \tilde{\sigma}^2} = -\frac{n}{2\tilde{\sigma}^2} + \frac{1}{2(\tilde{\sigma}^2)^2} \sum_{i=1}^n (y_i - \tilde{\beta}_0 - \tilde{\beta}_1 x_i)^2 = 0$$

SIMPLE LINEAR REGRESSION

- The MLEs are given by

$$\begin{aligned}\tilde{\beta}_0 &= \bar{y} - \tilde{\beta}_1 \bar{x} \\ \tilde{\beta}_1 &= \frac{\sum_{i=1}^n y_i (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \\ \tilde{\sigma}^2 &= \frac{\sum_{i=1}^n (y_i - \tilde{\beta}_0 - \tilde{\beta}_1 x_i)^2}{n}\end{aligned}$$

- MLEs $\tilde{\beta}_0$ and $\tilde{\beta}_1$ coincide with least squares estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ and are unbiased estimators of β_0 and β_1
- $\tilde{\sigma}^2$ and $\hat{\sigma}^2$ s.t. $\tilde{\sigma}^2 = \frac{n-1}{n} \hat{\sigma}^2 \Rightarrow \tilde{\sigma}^2$ biased estimator of σ^2 but asymptotically unbiased

SIMPLE LINEAR REGRESSION

- MLEs are consistent, i.e. they converge to the "true" values of the parameters asymptotically
- MLEs require stronger conditions w.r.t. least squares estimators, like Gaussianity and not only second moment knowledge
- $\hat{\beta}_0$ and $\hat{\beta}_1$ are best linear unbiased estimators since they have minimum variance among the unbiased estimators which are linear functions of y_i 's
 - Mean square error $MSE = E[(\hat{\theta} - \theta)^2]$ in estimating θ with $\hat{\theta}$

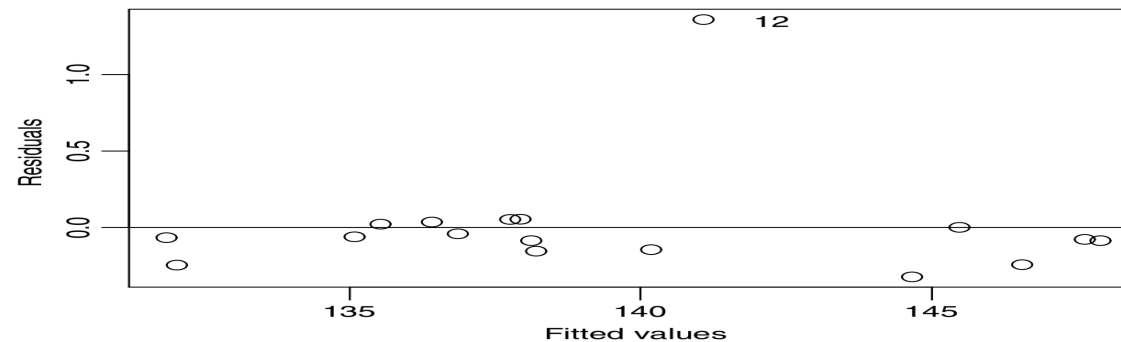
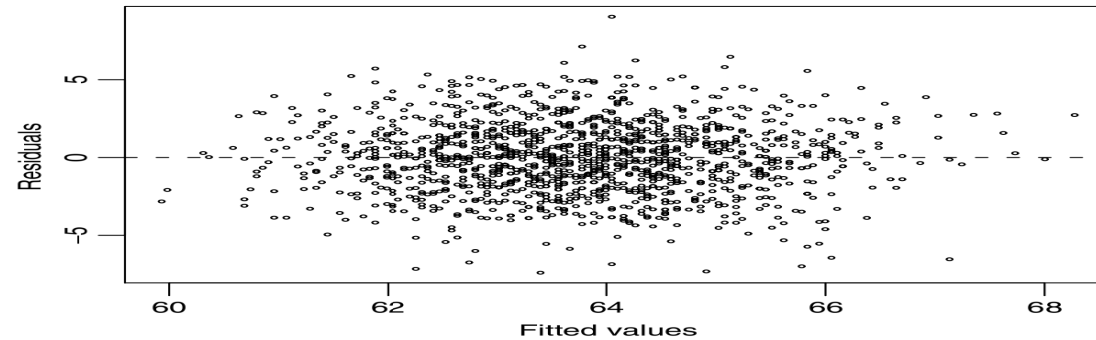
$$\begin{aligned} MSE &= E[(\hat{\theta} - E(\hat{\theta}) + E(\hat{\theta}) - \theta)^2] \\ &= E[(\hat{\theta} - E(\hat{\theta}))^2 + 2(\hat{\theta} - E(\hat{\theta}))(E(\hat{\theta}) - \theta) + (E(\hat{\theta}) - \theta)^2] \\ &= Var(\hat{\theta}) + 2\delta(\hat{\theta})E((\hat{\theta} - E(\hat{\theta}))) + [\delta(\hat{\theta})]^2 \\ &= Var(\hat{\theta}) + 2 * 0 + [\delta(\hat{\theta})]^2 \end{aligned}$$

- MSE minimised if unbiased, i.e. bias $\delta(\hat{\theta}) = 0$, and with minimum variance $\Rightarrow$ uniformly minimum variance unbiased estimators (UMVUE))

SIMPLE LINEAR REGRESSION

- Plots of residuals vs. other quantities are used to find failures of assumptions, especially w.r.t. fitted values
 - A null plot would indicate no failure of assumptions
 - Curvature might indicate that the fitted mean function is inappropriate
 - Residuals that seem to increase or decrease in average magnitude with the fitted values might indicate nonconstant residual variance
 - A few relatively large residuals may be indicative of outliers, cases for which the model is somehow inappropriate
- If there is no null plot, then actions are needed, like trying quadratic or nonparametric regression, different distribution for error or non common variance, or removal of outliers

SIMPLE LINEAR REGRESSION



- "Cloud" around 0 (top): example of good fit
- Outlier (bottom): is it a mistake, e.g., in typing data or an actual data?
- $\Rightarrow$ This needs investigation and possible repetition of the analysis without the outlier

SIMPLE LINEAR REGRESSION

Some further aspects

- $Y = \beta_0 + \beta_1 X^2 + \varepsilon$ linear model (in the parameters) replacing X^2 with Z
- $Y = \beta_0 + \beta_1^2 X + \varepsilon$ is not linear since the parameter is not linear
- In some books, $Var(\varepsilon_i) = \sigma^2/\omega_i \Rightarrow Var(y_i) = \sigma^2/\omega_i$, with ω_i a weight affecting the variability of the i -th observation
- The case above can be generalised to $\varepsilon_i \sim \mathcal{N}(0, \sigma_i^2)$ (heteroscedasticity)
- Furthermore, ε_i does not need to be Gaussian, but, e.g., Student
- Least square method $\Rightarrow$ Minimise $\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 = \sum_{i=1}^n |e_i|^2$
- LSE is in L_2 , but we could consider L_p (also with $p = 1$): minimise $\sum_{i=1}^n |e_i|^p$

AN EXAMPLE USING R

- Mean birthweight y (in kg) and gestational ages x (in weeks) of 1513 infants born to Caucasian mothers at St George's hospital, London, between 8/1982 and 3/1984

Gestational age (weeks)	Number of births	Birthweight means (kg)	Gestational age (weeks)	Number of births	Birthweight means (kg)
x_i	m_i	y_i	x_i	m_i	y_i
22	1	0.520	35	29	2.796
23	1	0.700	36	43	2.804
25	1	1.000	37	114	3.108
27	1	1.170	38	222	3.204
28	6	1.198	39	353	3.353
29	1	1.480	40	401	3.478
30	3	1.617	41	247	3.587
31	6	1.693	42	53	3.612
32	7	1.720	43	9	3.390
33	7	2.340	44	1	3.740
34	7	2.516			

- Install the package GLMsData and use data set `gestation*`

**From Dunn and Smyth. The authors claim the package is already in R but I found it neither in R nor in the list of available packages. I went to the CRAN website, downloaded the zip file and installed it locally*

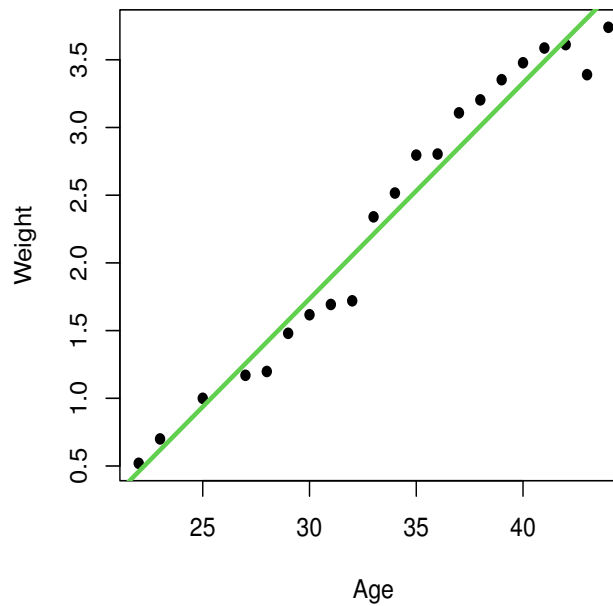
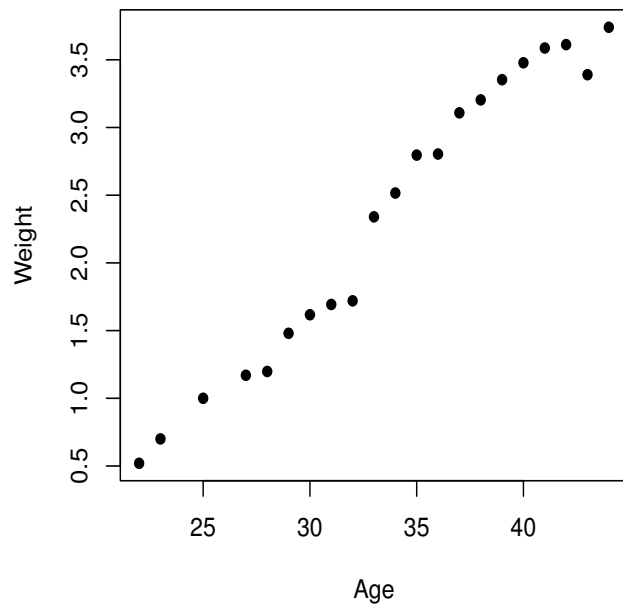
AN EXAMPLE USING R

We consider only two variables: Age (in weeks) and Weight (average weights of babies at each age), not the number of babies and the standard deviation (some are NA's because of just 1 baby) of the weights at each age

```
library(GLMsData); data(gestation)
head(gestation); attach(gestation)
summary(gestation)
plot(c(22,23,25,27:44),Weight,lwd=3,type='l',xlab="Age")
reg=lm(Weight ~ Age)
summary(reg)
plot(reg$res,lwd=3,pch=16,xlab="Age",ylab="Residual")
reg$coef
plot(c(22,23,25,27:44),Weight,lwd=3,pch=16,xlab="Age")
abline(reg$coef,lwd=3,col=3)
```

AN EXAMPLE USING R

- $\hat{\beta}_0 = -3.0499$, $\hat{\beta}_1 = 0.1595$, $R^2 = 0.9668$



BAYES THEOREM

- Patient subject to medical diagnostic test (P or N) for a disease D
- *Sensitivity* .95, i.e. $\mathbb{P}(P|D) = .95$
- *Specificity* .9, i.e. $\mathbb{P}(P^C|D^C) = \mathbb{P}(N|D^C) = .9$
- Physician's belief on patient having the disease 1%, i.e. $\mathbb{P}(D) = .01$
 - Knowledge about **that** patient
 - Knowledge about people with similar characteristics (age, gender, etc.)
 - Knowledge about the population in an area
 - Other sources of knowledge or uninformative guess
- Positive test $\Rightarrow \mathbb{P}(D|P)$?

BAYES THEOREM

$$\begin{aligned}\mathbb{P}(D|P) &= \frac{\mathbb{P}(D \cap P)}{\mathbb{P}(P)} = \frac{\mathbb{P}(P|D)\mathbb{P}(D)}{\mathbb{P}(P|D)\mathbb{P}(D) + \mathbb{P}(P|D^C)\mathbb{P}(D^C)} \\ &= \frac{.95 \cdot .01}{.95 \cdot .01 + .1 \cdot .99} = .0875\end{aligned}$$

Positive test updates belief on patient having the disease:
from 1% to 8.75%

Prior opinion updated into posterior one

If $\mathbb{P}(D) = .1 \Rightarrow \mathbb{P}(D|P) = .5135$

If $\mathbb{P}(D) = .2 \Rightarrow \mathbb{P}(D|P) = .7037$

BAYES THEOREM

- Partition $\{A_1, \dots, A_n\}$ of Ω and $B \subset \Omega : \mathbb{P}(B) > 0$

$$\mathbb{P}(A_i|B) = \frac{\mathbb{P}(B|A_i)P(A_i)}{\sum_{j=1}^n \mathbb{P}(B|A_j)P(A_j)}$$

- X r.v. with density $f(x|\lambda)$, prior $\pi(\lambda)$

$$\Rightarrow \text{posterior } \pi(\lambda|x) = \frac{f(x|\lambda)\pi(\lambda)}{\int f(x|\omega)\pi(\omega)d\omega}$$

ILLUSTRATIVE EXAMPLE: BAYESIAN APPROACH

Light bulb lifetime $\Rightarrow X \sim \mathcal{E}(\lambda)$ & $f(x; \lambda) = \lambda e^{-\lambda x}$ $x, \lambda > 0$

- Sample $\underline{X} = (X_1, \dots, X_n)$, i.i.d. $\mathcal{E}(\lambda)$
- Likelihood $l_x(\lambda) = \prod_{i=1}^n f(X_i; \lambda) = \lambda^n e^{-\lambda \sum_{i=1}^n X_i}$
- Prior $\lambda \sim \mathcal{G}(\alpha, \beta)$, $\pi(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta \lambda}$
- Posterior $\pi(\lambda|\underline{X}) \propto \lambda^n e^{-\lambda \sum_{i=1}^n X_i} \cdot \lambda^{\alpha-1} e^{-\beta \lambda}$
 $\Rightarrow \lambda|\underline{X} \sim \mathcal{G}(\alpha + n, \beta + \sum_{i=1}^n X_i)$

Posterior distribution fundamental in Bayesian analysis

CONJUGATE PRIORS

- We just saw that a gamma prior on the parameter of an exponential model leads to a gamma posterior
- $\Rightarrow$ The gamma distribution is a conjugate prior for the exponential model
- Does conjugacy occur always? Unfortunately not and simulation methods, e.g. MCMC (Markov chain Monte Carlo), are needed to get samples from the posterior distribution
- There are some relevant cases of conjugacy and we will see some of them:
 - Beta prior conjugate w.r.t. Bernoulli, binomial, geometric models
 - Dirichlet prior conjugate w.r.t. multinomial model
 - Gamma prior conjugate w.r.t. exponential, Poisson models
 - Gaussian prior conjugate w.r.t. Gaussian model with fixed variance/covariance matrix and unknown mean
 - Gaussian-Inverse gamma prior w.r.t. univariate Gaussian model with unknown mean and variance

CONJUGATE PRIOR FOR GAUSSIAN

- $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$
- Mean/median $\mu \in \Re$ and variance $\sigma^2 > 0$ unknown
- $\underline{X} = (X_1, \dots, X_n)$
- Conjugate normal-inverse gamma prior
- Prior $\pi(\mu, \sigma^2) = \pi(\mu|\sigma^2)\pi(\sigma^2)$
- $\mu|\sigma^2 \sim \mathcal{N}(\mu_0, \tau^2\sigma^2)$
- $\sigma^2 \sim \mathcal{IG}(\alpha, \beta)$ Inverse gamma
- $\pi(\sigma^2) = \frac{\beta^\alpha}{\Gamma(\alpha)}(\sigma^2)^{-\alpha-1}e^{-\beta/\sigma^2}$, with $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1}e^{-x}dx$

CONJUGATE PRIOR FOR GAUSSIAN

- After some computations (left as an exercise) we get the posterior $\pi(\mu, \sigma^2 | \underline{X}) = \pi(\mu | \sigma^2, \underline{X}) \pi(\sigma^2 | \underline{X})$ with

$$- \mu | \sigma^2, \underline{X} \sim \mathcal{N} \left(\frac{\sum_{i=1}^n X_i + \mu_0 / \tau^2}{n + 1 / \tau^2}, \frac{\sigma^2}{n + 1 / \tau^2} \right)$$

$$- \sigma^2 | \underline{X} \sim \mathcal{IG}(\alpha + (n + 1)/2, \beta)$$

- The posterior marginal of μ , i.e. $\pi(\mu | \underline{X})$, has a Student- t distribution

$$• \sigma^2 | \mu, \underline{X} \sim \mathcal{IG}(\alpha + (n + 1)/2, \beta + \sum_{i=1}^n (X_i - \mu)^2 / 2 + (\mu - \mu_0)^2 / (2\tau^2))$$

- $\Rightarrow$ useful for MCMC (Gibbs sampling)

PARAMETER ESTIMATION - DECISION ANALYSIS

- Loss function $L(\lambda, a)$, $a \in \mathcal{A}$ action space
- Minimize $\mathcal{E}^{\pi(\lambda|\underline{X})} L(\lambda, a) = \int L(\lambda, a) \pi(\lambda|\underline{X}) d\lambda$ w.r.t. a
 - $\Rightarrow \hat{\lambda}$ Bayesian optimal estimator of λ
 - $\hat{\lambda}$ posterior median if $L(\lambda, a) = |\lambda - a|$
 - $\hat{\lambda}$ posterior mean $\mathcal{E}^{\pi(\lambda|\underline{X})} \lambda$ if $L(\lambda, a) = (\lambda - a)^2$

$$\begin{aligned}\mathcal{E}^{\pi(\lambda|\underline{X})} L(\lambda, a) &= \int (\lambda - a)^2 \pi(\lambda|\underline{X}) d\lambda \\ &= \int \lambda^2 \pi(\lambda|\underline{X}) d\lambda - 2a \int \lambda \pi(\lambda|\underline{X}) d\lambda + a^2 \cdot 1 \\ &= \int \lambda^2 \pi(\lambda|\underline{X}) d\lambda - 2a \mathcal{E}^{\pi(\lambda|\underline{X})} \lambda + a^2\end{aligned}$$

CREDIBLE INTERVALS

- In Bayesian statistics the parameter λ is considered a r.v. and it is possible to compute the posterior probability $\mathcal{P}(\lambda \in A|\underline{X})$ for a measurable set A
- $\Rightarrow$ Credible set, as a counterpart of the frequentist confidence set, but with very different meaning
- If the set is an interval, then we call it *credible interval at 100y%*, if its posterior probability is y
- We are interested also in the *highest posterior density (HPD) sets*, which are the ones with the smallest Lebesgue measure among those with a given posterior probability
- Light bulb: $\mathcal{P}(\lambda \leq z|\underline{X}) = \int_0^z \frac{(\beta + \sum X_i)^{\alpha+n}}{\Gamma(\alpha+n)} \lambda^{\alpha+n-1} e^{-(\beta + \sum X_i)\lambda} d\lambda$

CREDIBLE INTERVALS

- One observation $X \sim \mathcal{N}(\mu, 1)$
- Prior $\mu \sim \mathcal{N}(0, 1)$
- Posterior
$$\pi(\mu|x) \propto e^{-(x-\mu)^2/2} \cdot e^{-\mu^2/2} \propto e^{-(\mu^2 - x\mu)} \propto \exp \frac{1}{2 \cdot 1/2} (\mu - x/2)^2$$
$$\Rightarrow \mu|x \sim \mathcal{N}(x/2, 1/2)$$
- $Z = \frac{\mu - x/2}{\sqrt{1/2}} \sim \mathcal{N}(0, 1)$
- Quantiles $Z_{.975} = 1.96$ and $Z_{.025} = -1.96$
- $\Rightarrow P(Z_{.025} \leq Z \leq Z_{.975}) = \left(-1.96 \leq \frac{\mu - x/2}{\sqrt{1/2}} \leq 1.96 \right) = .95$
- $\Rightarrow \left(x/2 - 1.96\sqrt{1/2}, x/2 + 1.96\sqrt{1/2} \right)$ credible interval at 95%

PREDICTION

- After observing an i.i.d. sample $\underline{X} = (X_1, \dots, X_n)$, what can we say about a next observation X_{n+1} from the same density $f(X|\lambda)$?
- We could consider the next observations $X_{n+1}, \dots, X_{n+j}$ but we take $j = 1$ for simplicity
- When considering observations over time we prefer to use the term *forecast* instead of *prediction* (e.g., weather forecast)
- Given the sample $\underline{X}$ and the prior $\pi(\lambda)$, then the posterior $\pi(\lambda|\underline{X})$ is used to compute the posterior predictive density (absolutely continuous case here) for X_{n+1}
$$f(X_{n+1}|\underline{X}) = \int f(X_{n+1}|\lambda, \underline{X})\pi(\lambda|\underline{X})d\lambda = \int f(X_{n+1}|\lambda)\pi(\lambda|\underline{X})d\lambda$$
- Prior predictive densities can be used to compare model via Bayes factor (more later)
- Posterior predictive densities can be used to assess the goodness of fit of a model through the prediction error, using part of the data to get the posterior and the remaining one to get predicted values (e.g. predicted posterior mean/median) and compare them with actual ones

PREDICTION

- Light bulb: $X_{n+1}|\lambda \sim \mathcal{E}(\lambda)$, $\lambda|\underline{X} \sim \mathcal{G}(\alpha + n, \beta + \sum X_i)$
- Posterior predictive density for X_{n+1}

$$\begin{aligned} f_{X_{n+1}}(X_{n+1}|\underline{X}) &= \int_0^\infty \lambda e^{-\lambda X_{n+1}} \cdot \frac{(\beta + \sum X_i)^{\alpha+n}}{\Gamma(\alpha + n)} \lambda^{\alpha+n-1} e^{-\lambda(\beta + \sum X_i)} d\lambda \\ &= \frac{(\beta + \sum X_i)^{\alpha+n}}{\Gamma(\alpha + n)} \int_0^\infty \lambda^{\alpha+n+1-1} e^{-\lambda(\beta + \sum X_i + X_{n+1})} d\lambda \\ &= \frac{(\beta + \sum X_i)^{\alpha+n}}{\Gamma(\alpha + n)} \frac{\Gamma(\alpha + n + 1)}{(\beta + \sum X_i + X_{n+1})^{\alpha+n+1}} \\ &= (\alpha + n) \frac{(\beta + \sum X_i)^{\alpha+n}}{(\beta + \sum X_i + X_{n+1})^{\alpha+n+1}} \end{aligned}$$

- I found first the constant knowing that the density integrates to 1 and then I used the property $\Gamma(n + 1) = n\Gamma(n)$

MARKOV CHAIN MONTE CARLO

- A Markov chain is a sequence of r.v.'s $\{X_n\}$ such that the distribution of any X_n depends on the past only through X_{n-1}
- $\mathbb{P}(X_n | X_{n-1}, X_{n-2}, \dots, X_1) = \mathbb{P}(X_n | X_{n-1}), \forall n$
- MCMC used to draw samples "converging" towards posterior $\pi(\theta | \underline{X})$
- Name MCMC due to simulations based on transition distributions $p(\theta^i | \theta^{i-1})$
- Many works dealt with the theory justifying MCMC, ensuring theoretical convergence to the posterior distribution: we will not discuss them except for mentioning that the posterior distribution is the stationary distribution of an appropriate Markov chain
- Many works addressed the issue of empirically guaranteeing the practical convergence: we will discuss them briefly
- Many MCMC methods: here only Gibbs sampling and Metropolis-Hastings algorithm
- No mention of other simulation methods, like Variational Bayes and Approximate Bayesian Computation

MARKOV CHAIN MONTE CARLO

- Sample $\underline{X}$ and parameter $\theta = (\theta_1, \dots, \theta_n)$
- Notation $\theta_{-i} = (\theta_1, \dots, \theta_{i-1}, \theta_{i+1}, \dots, \theta_n)$, for $i = 1, \dots, n$
- *Gibbs sampling* is used when $\pi(\theta|\underline{X})$ is not available but all $\pi(\theta_i|\theta_{-i}, \underline{X})$, $i = 1, \dots, n$, are
- Example seen earlier: $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ with $\mu \in \Re$ and $\sigma^2 > 0$ unknown
 - Prior $\pi(\mu, \sigma^2) = \pi(\mu|\sigma^2)\pi(\sigma^2)$
 - $\mu|\sigma^2 \sim \mathcal{N}(\mu_0, \tau^2\sigma^2)$
 - $\sigma^2 \sim \mathcal{IG}(\alpha, \beta)$ Inverse gamma
 - $\mu|\sigma^2, \underline{X} \sim \mathcal{N}\left(\frac{\sum_{i=1}^n X_i + \mu_0/\tau^2}{n + 1/\tau^2}, \frac{\sigma^2}{n + 1/\tau^2}\right)$
 - $\sigma^2|\mu, \underline{X} \sim \mathcal{IG}(\alpha + (n + 1)/2, \beta + \sum_{i=1}^n (X_i - \mu)^2/2 + (\mu - \mu_0)^2/(2\tau^2))$

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- In words, Gibbs sampling consists of a "sufficient" number of steps in which each parameter θ_i is sequentially drawn from its *full conditional distribution* $\pi(\theta_i|\theta_{-i}, \underline{X})$, where θ_{-i} contains the values of $\theta_1, \dots, \theta_{i-1}$ generated at the current step and those of $\theta_{i+1}, \dots, \theta_n$ generated at the previous step
- Algorithm
 1. Set $\theta^{(0)} = (\theta_1^{(0)}, \dots, \theta_n^{(0)})$ and $j = 0$
 2. Set $j = j + 1$
 3. For $i = 1, \dots, n$, draw $\theta_i^{(j)}$ from $\pi(\theta_i|\theta_1^{(j)}, \dots, \theta_{i-1}^{(j)}, \theta_{i+1}^{(j-1)}, \dots, \theta_n^{(j-1)}, \underline{X})$
 4. If $j < N$ (set a priori) then go back to (2)
 5. $\Rightarrow \theta^{(j)}, j = 1, \dots, N$, used to get a sample from the posterior distribution
- Some $\theta^{(j)}$'s might be discarded, e.g. initial ones (more later)

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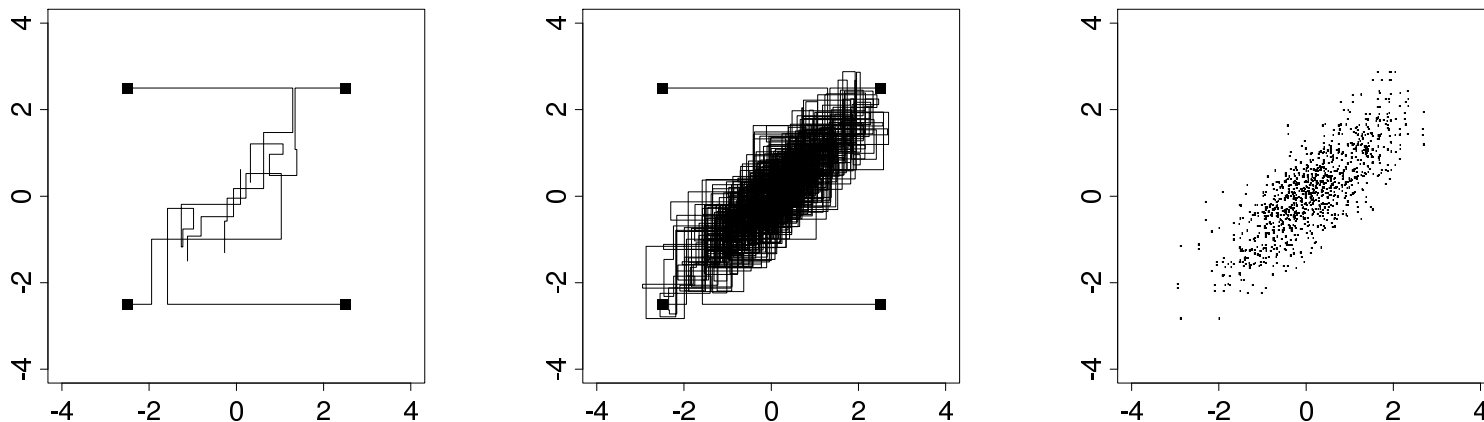
- Consider a single observation (y_1, y_2) from a bivariate Gaussian with unknown mean $\theta = (\theta_1, \theta_2)$ and known covariance matrix $\begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$
- Uniform prior on θ : $\pi(\theta) \propto c, c > 0$
- $\Rightarrow$ Posterior $\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} | y \sim \mathcal{N} \left(\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right)$
- Although it is simple to draw directly from the joint posterior distribution of (θ_1, θ_2) , for the purpose of exposition we demonstrate the Gibbs sampler here
- Simulate (alternating) from known full conditional distributions
 - $\theta_1 | \theta_2, y \sim \mathcal{N}(y_1 + \rho(\theta_2 - y_2), 1 - \rho^2)$
 - $\theta_2 | \theta_1, y \sim \mathcal{N}(y_2 + \rho(\theta_1 - y_1), 1 - \rho^2)$

*Example from Gelman et al., *Bayesian Data Analysis, Third Edition*, freely available at <http://www.stat.columbia.edu/~gelman/book/BDA3.pdf>

MARKOV CHAIN MONTE CARLO

- Take $\rho = 0.8$ and $(y_1, y_2) = (0, 0)$
- $\Rightarrow$ Posterior $\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} | y \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix} \right)$
- Four independent sequences starting at $(\pm 2.5, \pm 2.5)$ to remove dependence on initial point
- Sequences run until convergence to the posterior is achieved (more later on checking for convergence)
- By convergence we mean that the drawn samples are from an approximating distribution close to the posterior one (our target)
- Use of just part of the data, removing the initial ones since they might not be in the approximating distribution (this operation is called *burn-in*)
- Sometimes one aims to reduce correlation between samples so that just 1 every m is kept

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- Left: First 10 iterations for four independent sequences starting at $(\pm 2.5, \pm 2.5)$
- Center: After 500 iterations, the sequences have reached approximate convergence
- Right: The points from the second halves of the sequences, discarding the first 250 samples values of each sequence (burn-in)
- Often just one sequence is drawn but for longer time
- Note how the samples are around $(0, 0)$ and showing a strong positive correlation, as expected knowing the exact joint posterior

MARKOV CHAIN MONTE CARLO

- In Gibbs sampling we assumed that it was always possible to get the full conditional $\pi(\theta_i|\theta_{-i}, \underline{X})$ for all i 's but is not always the case
- Sometimes we know only $\pi(\theta_i|\theta_{-i}, \underline{X}) \propto q(\theta_i|\theta_{-i}, \underline{X})$ where $q(\cdot)$ is not a density function
- It is a similar case to what seen before when we considered $q(\lambda|y) = f(y|\lambda)\pi(\lambda)$ known, unlike its integral w.r.t. λ which is the normalising constant in Bayes Theorem
- In this case we will use *Metropolis-Hastings steps within Gibbs*
- The Metropolis-Hastings algorithm allows to draw a value θ_i^* from a proposal density $p(\theta_i)$ and accept either it or $\theta_i^{(j-1)}$ as $\theta_i^{(j)}$ with probabilities depending on both p and q
- The proposal density for θ_i^* could be chosen, e.g., either as the same for each iteration or as dependent on the previous $\theta_i^{(j-1)}$

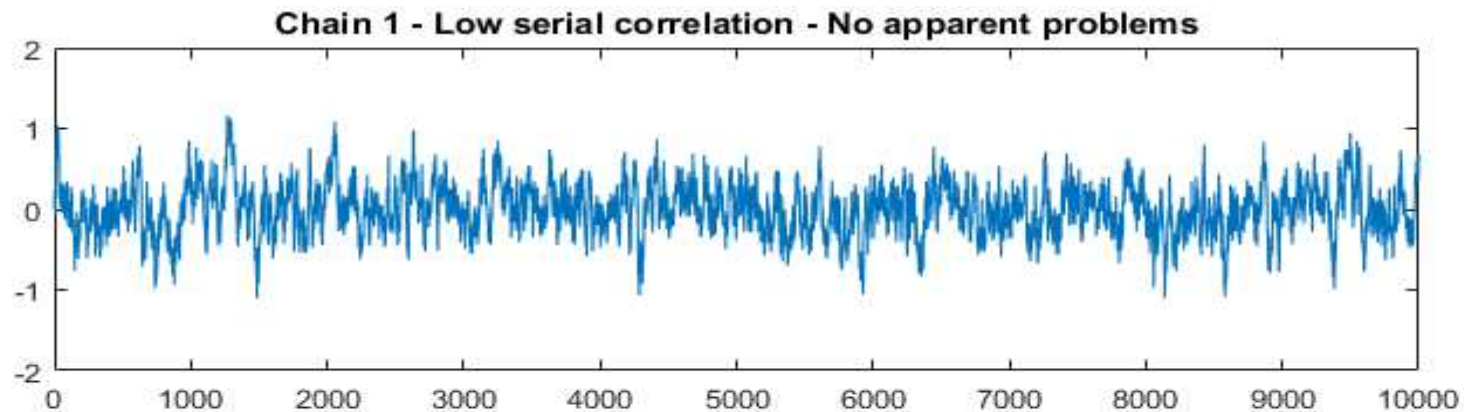
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- For simplicity of notation, let us remove the index i and the conditioning on the other parameters and consider just θ
- Suppose that $\pi(\theta|y)$ is known just up to a constant, i.e. $\pi(\theta|y) \propto q(\theta|y)$, or
$$\pi(\theta|y) = \frac{q(\theta|y)}{\int q(\theta|y)d\theta}$$
- We start with an initial value $\theta^{(0)}$ s.t. $\pi(\theta^{(0)}|y) > 0$
- For each iteration $j = 1, \dots, N$, generate a proposal θ^* from a proposal density $p_j(\theta|\theta^{(j-1)})$
- Compute the ratio $r = \frac{q(\theta^*|y)/p(\theta^*|\theta^{(j-1)})}{q(\theta^{(j-1)}|y)/p(\theta^{(j-1)}|\theta^*)}$
- Set $\theta^{(j)} = \begin{cases} \theta^* & \text{with probability } \min(r, 1) \\ \theta^{(j-1)} & \text{otherwise} \end{cases}$

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- We already saw that running more than one simulation at the time and removing the initial values should reduce the dependence on the initial values
- The proposal distributions are often chosen depending on the value at the previous iteration, e.g. a Gaussian distribution centered at it, or independently from it, possibly the same at all iterations, e.g. Gaussians with the same mean
- Many tools developed to check convergence of the sequence to the true distribution
- The simplest, graphical, tool to assess convergence is to check if the plot of the sample mean stabilises as the iterations grow (if not, then no convergence)
- Given a sample $\theta^{(S+1)}, \dots, \theta^{(N)}$, with a burn-in of size S , then estimators of $E(h(\theta)|y)$ are given by $\frac{\sum_{j=S+1}^N h(\theta^{(j)})}{N-S}$, like
 - $E(\theta|y) \approx \frac{\sum_{j=S+1}^N \theta^{(j)}}{N-S}$
 - $\mathbb{P}(\theta \in A|y) \approx \frac{\#\{\theta^{(j)} \in A\}_{j=S+1}^N}{N-S}$

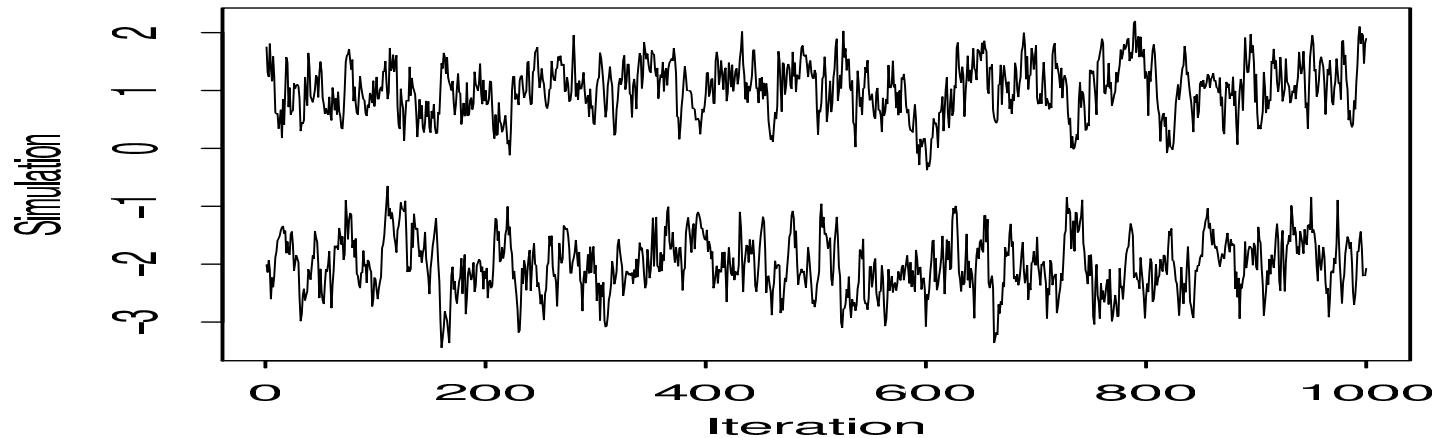
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- Trace plots are heuristic tools, widely used to check convergence of the MCMC
- They plot the values of each parameter for all the iterations
- They are "good" when the plot keeps jumping within a set which denotes where the posterior density is concentrated
- The trace plot in the figure is a good one, unlike the next ones

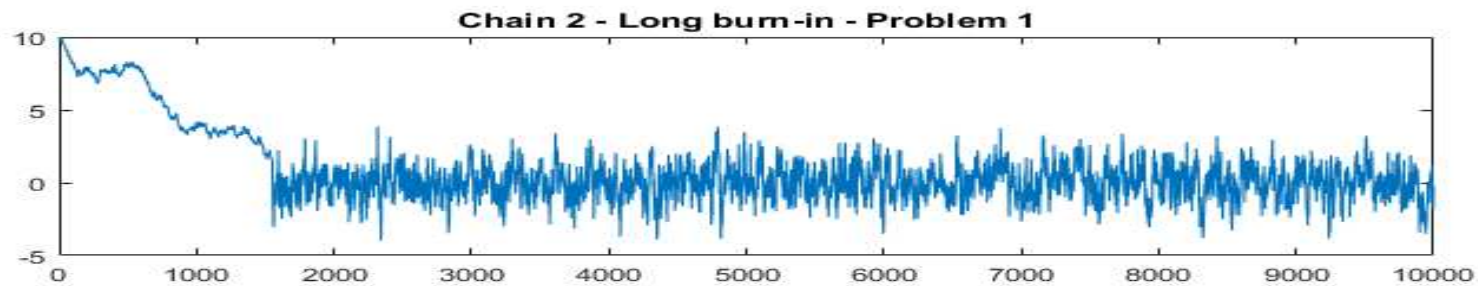
*Plots from www.statlect.com

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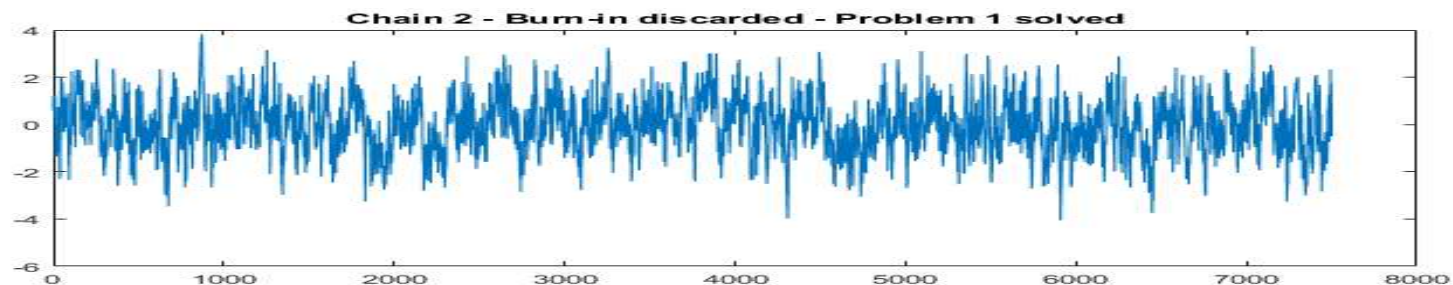


- Here two sequences have been running and both of them are converging but to two different values
- In general, a plot like this is not desirable since it does not give a clear indication about where the posterior density is, unless the density is bimodal
- In the latter case one would expect the chain to jump from one mode to another

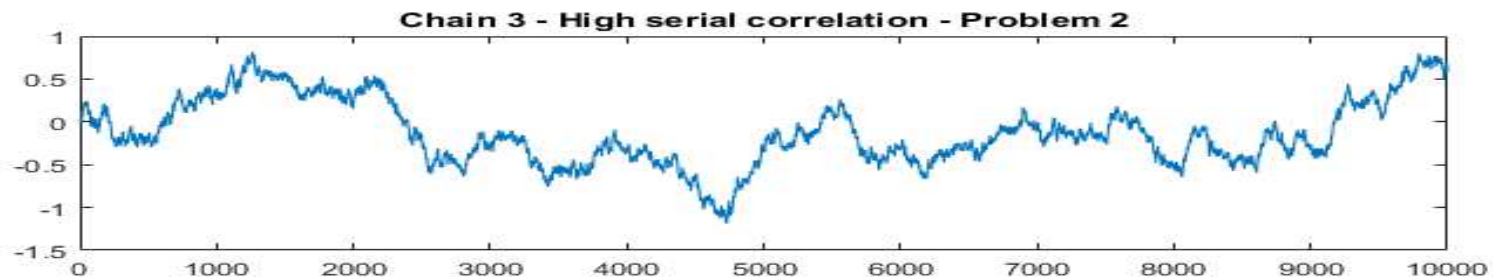
MARKOV CHAIN MONTE CARLO



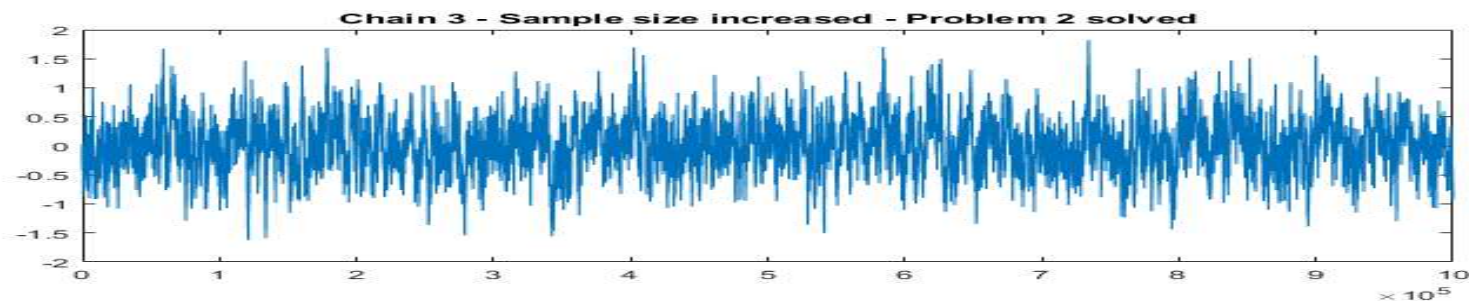
- The first part of the sample looks very different from the remaining part.
- Most likely, the initial distribution and the distributions of the subsequent terms of the chain were very different from the target distribution, but then the chain slowly converged to the target distribution
- The problem can be solved by removing the initial values (burn-in)



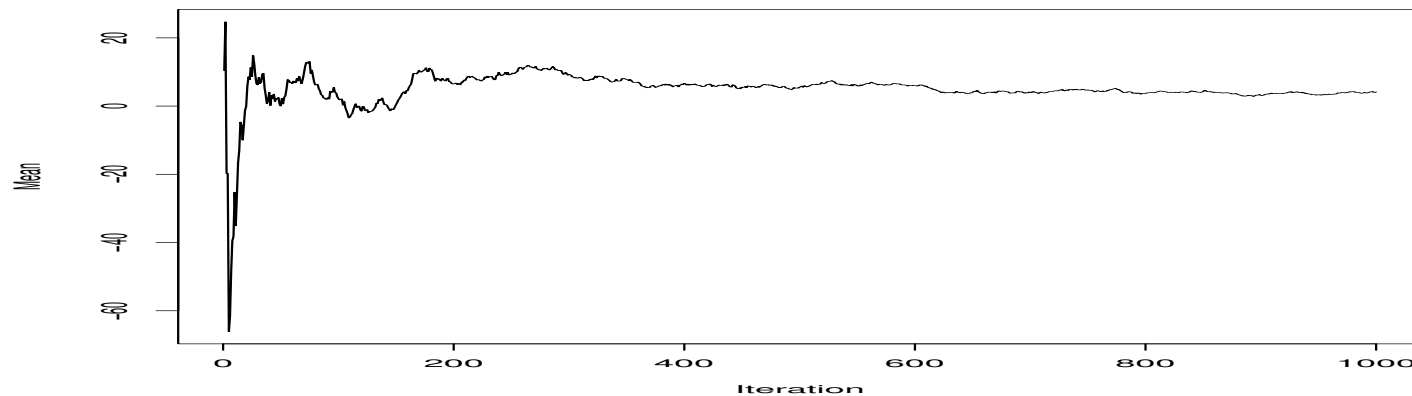
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- A lot of autocorrelation between the draws ($\Rightarrow$ lack of independence)
- Chain very slow in exploring the sample space, explored only few times
- The problem could be due to a small number of iterations $\Rightarrow$ run longer and, possibly, take one draw out of m to avoid large sample size and remove autocorrelation



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- We can also show how estimators, like posterior mean, are evolving as the number of iterations increases and if they are stabilising around a value
- In this case the plot denotes the evolution of the sample mean, estimator of the parameter, as the iterations increase

REGRESSION

- Observations: $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, i = 1, \dots, n$
- X_i 's are supposed known here but they could be r.v.'s as well
- We assume that $\varepsilon_1, \dots, \varepsilon_n$ are i.i.d. $\mathcal{N}(0, \sigma^2)$
- Notation: $\underline{Y} = (Y_1, \dots, Y_n)$ and $\underline{X} = (X_1, \dots, X_n)$
- Likelihood function $L(\beta_0, \beta_1, \sigma^2 | \underline{Y}, \underline{X})$ given by

$$\begin{aligned} \prod_{i=1}^n f(Y_i | X_i, \beta_0, \beta_1, \sigma^2) &= \prod_{i=1}^n \left\{ \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(Y_i - \beta_0 - \beta_1 X_i)^2}{2\sigma^2}\right\} \right\} \\ &\propto \frac{1}{(\sigma^2)^{n/2}} \exp\left\{-\frac{\sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2}{2\sigma^2}\right\} \end{aligned}$$

- Independent priors with known hyperparameters:
 $\beta_0 \sim \mathcal{N}(0, \tau_1^2), \beta_1 \sim \mathcal{N}(0, \tau_2^2)$ and $\sigma^2 \sim \mathcal{IG}(a, b)$

REGRESSION

- Posterior distribution

$$\begin{aligned} \pi(\beta_0, \beta_1, \sigma^2 | \underline{Y}, \underline{X}) &\propto \frac{1}{(\sigma^2)^{n/2}} \exp\left\{-\frac{\sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2}{2\sigma^2}\right\} \cdot \\ &\cdot \exp\{-\beta_0^2/(2\tau_1^2)\} \exp\{-\beta_1^2/(2\tau_2^2)\} \frac{1}{(\sigma^2)^{a+1}} \exp\{-b/\sigma^2\} \end{aligned}$$

- Conditional on β_0 : $\beta_0 | \beta_1, \sigma^2, \underline{Y}, \underline{X} \sim \mathcal{N}\left(\frac{\sum_{i=1}^n (Y_i - \beta_1 X_i)}{n + \sigma^2/\tau_1^2}, \frac{1}{n/\sigma^2 + 1/\tau_1^2}\right)$

$$\begin{aligned} \pi(\beta_0 | \beta_1, \sigma^2, \underline{Y}, \underline{X}) &\propto \exp\left\{-\frac{(n\beta_0^2 - 2\beta_0 \sum_{i=1}^n (Y_i - \beta_1 X_i))}{2\sigma^2}\right\} \exp\{-\beta_0^2/(2\tau_1^2)\} \\ &\propto \exp\left\{-\frac{1}{2} \left[\left(\frac{n}{\sigma^2} + \frac{1}{\tau_1^2} \right) \beta_0^2 - 2 \frac{\beta_0}{\sigma^2} \sum_{i=1}^n (Y_i - \beta_1 X_i) \right] \right\} \\ &\propto \exp\left\{-\frac{1}{2(n/\sigma^2 + 1/\tau_1^2)^{-1}} \left[\beta_0^2 - 2 \frac{\beta_0}{\sigma^2} \frac{\sum_{i=1}^n (Y_i - \beta_1 X_i)}{n/\sigma^2 + 1/\tau_1^2} \right] \right\} \end{aligned}$$

REGRESSION

- Conditional on β_1 : $\beta_1 | \beta_0, \sigma^2, \underline{Y}, \underline{X} \sim \mathcal{N} \left(\frac{\sum_{i=1}^n X_i (Y_i - \beta_0)}{\sum_{i=1}^n X_i^2 + \sigma^2 / \tau_1^2}, \frac{1}{\sum_{i=1}^n X_i^2 / \sigma^2 + 1 / \tau_1^2} \right)$

$$\begin{aligned} \pi(\beta_1 | \beta_0, \sigma^2, \underline{Y}, \underline{X}) &\propto \exp \left\{ -\frac{\beta_1^2 \sum_{i=1}^n X_i^2 - 2\beta_1 \sum_{i=1}^n X_i (Y_i - \beta_0)}{2\sigma^2} \right\} \exp \{ -\beta_1^2 / (2\tau_1^2) \} \\ &\propto \exp \left\{ -\frac{1}{2} \left[\left(\frac{\sum_{i=1}^n X_i^2}{\sigma^2} + \frac{1}{\tau_1^2} \right) \beta_1^2 - 2 \frac{\beta_1}{\sigma^2} \sum_{i=1}^n X_i (Y_i - \beta_0) \right] \right\} \\ &\propto \exp \left\{ -\frac{1}{2 \left(\sum_{i=1}^n \frac{X_i^2}{\sigma^2} + \frac{1}{\tau_1^2} \right)^{-1}} \left[\beta_1^2 - 2 \frac{\beta_1}{\sigma^2} \frac{\sum_{i=1}^n X_i (Y_i - \beta_0)}{\sum_{i=1}^n X_i^2 / \sigma^2 + 1 / \tau_1^2} \right] \right\} \end{aligned}$$

- Conditional on σ^2 : $\sigma^2 | \beta_0, \beta_1, \underline{Y}, \underline{X} \sim \mathcal{IG} \left(a + n/2, b + \frac{\sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2}{2} \right)$

$$\pi(\sigma^2 | \beta_0, \beta_1, \underline{Y}, \underline{X}) \propto \frac{1}{(\sigma^2)^{n/2}} \exp \left\{ -\frac{\sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2}{2\sigma^2} \right\} \frac{1}{(\sigma^2)^{a+1}} \exp \{ -b / \sigma^2 \}$$

- $\Rightarrow$ Gibbs sampling

REGRESSION

- Different approach, with slight change, from Press, J. (2002), *Subjective and Objective Bayesian Statistics*, Wiley

- $\pi(\beta_0, \beta_1, \sigma^2) \propto 1/\sigma^2$

- Joint posterior

$$\pi(\beta_0, \beta_1, \sigma^2 | \underline{Y}, \underline{X}) \propto \frac{1}{(\sigma^2)^{n/2}} \exp\left\{-\frac{\sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2}{2\sigma^2}\right\} \frac{1}{\sigma^2}$$

- Integrating out $\sigma^2 \Rightarrow (\beta_0, \beta_1) | \underline{Y}, \underline{X}$ bivariate Student distribution

$$\pi(\beta_0, \beta_1 | \underline{Y}, \underline{X}) \propto \frac{1}{\left[\sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2\right]^{n/2}}$$

- Bivariate Student's $t \Rightarrow$ Marginal Student's t

- $\Rightarrow \beta_0 | \underline{Y}, \underline{X} \sim t, \beta_1 | \underline{Y}, \underline{X} \sim t$ and $\sigma^2 | \beta_0, \beta_1, \underline{Y}, \underline{X} \sim \mathcal{IG}$ as before

- $\sigma_2^{(j)}$ from inverse gamma with values $\beta_0^{(j)}$ and $\beta_1^{(j)}$ generated from the t-distributions

REGRESSION

- Another different approach from Cowles, M.K., (2013), *Applied Bayesian Statistics*, Springer
- Center the X_i 's around their mean $\bar{X} \Rightarrow X_i \rightarrow X_i - \bar{X}$
- $\Rightarrow Y_i|X_i, \beta_0, \beta_1, \sigma^2 \sim \mathcal{N}(\beta_0 + \beta_1(X_i - \bar{X}), \sigma^2), i = 1, \dots, n$
- $\pi(\beta_0, \beta_1, \sigma^2) \propto 1/\sigma^2$
- We will consider three sufficient statistics: $\hat{\beta}_1, \hat{\beta}_2, SSR$ (sum of squared residuals)
- Given a r.v. X with density $f(X|\theta)$, a statistic $t = T(X)$ is said sufficient for θ if $f(X|t = T(X))$ does not depend on θ
- In words, a sufficient statistic contains all the information provided by the data about the model parameters
- Those statistics are estimators from a frequentist viewpoint

REGRESSION

- Likelihood: $\prod_{i=1}^n \left\{ \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{1}{2\sigma^2} [Y_i - \beta_0 - \beta_1(X_i - \bar{X})]^2\right\} \right\}$ i.e.

$$\frac{1}{[\sqrt{2\pi}\sigma]^n} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n [Y_i - \beta_0 - \beta_1(X_i - \bar{X})]^2\right\}$$
- Loglikelihood $\propto l(\beta_0, \beta_1, \sigma^2) = -\frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n [Y_i - \beta_0 - \beta_1(X_i - \bar{X})]^2$
- $\frac{\partial l}{\partial \beta_0} = \frac{1}{\sigma^2} \sum_{i=1}^n [Y_i - \beta_0 - \beta_1(X_i - \bar{X})] = 0 \Leftrightarrow \sum_{i=1}^n Y_i - n\beta_0 - \beta_1 \sum_{i=1}^n [X_i - \bar{X}] = 0$
- $\Rightarrow \hat{\beta}_0 = \frac{\sum_{i=1}^n Y_i}{n} = \bar{Y}$ since $\sum_{i=1}^n [X_i - \bar{X}] = 0$
- Note: if we had not centered the X_i around $\bar{X}$, we would have got $\hat{\beta}_1 = \bar{Y} - \hat{\beta}_1 \bar{X}$

REGRESSION

$$\begin{aligned}\frac{\partial l}{\partial \beta_1} &= \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X}) [Y_i - \beta_0 - \beta_1(X_i - \bar{X})] = 0 \\ \Leftrightarrow \sum_{i=1}^n (X_i - \bar{X})(Y_i - \hat{\beta}_1) - \hat{\beta}_2 \sum_{i=1}^n (X_i - \bar{X})^2 &= 0 \\ \Leftrightarrow \hat{\beta}_2 &= \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}\end{aligned}$$

- We have plugged-in $\hat{\beta}_1$ and $\hat{\beta}_2$ since the equality conditions are to be satisfied by them
- Sum of squared residuals $SSR = \sum_{i=1}^n [Y_i - \hat{\beta}_1 - \hat{\beta}_2(X_i - \bar{X})]^2$
- Remember: in simple linear regression $\Rightarrow$ Sample variance $S^2 = \frac{SSR}{n-2}$
- We now go back to the Bayesian computations, using $\hat{\beta}_1, \hat{\beta}_2$ and SSR

REGRESSION

- Priors: $\pi(\beta_0) \propto c_1, \pi(\beta_1) \propto c_2, \sigma^2 \propto 1/\sigma^2$
- $\Rightarrow \pi(\beta_0, \beta_1, \sigma^2) \propto 1/\sigma^2$
- Posterior $\pi(\beta_0, \beta_1, \sigma^2 | \underline{X}, \underline{Y})$

$$\begin{aligned} &\propto \frac{1}{(\sigma^2)^{n/2}} \exp\left\{-\frac{\sum_{i=1}^n [Y_i - \beta_0 - \beta_1(X_i - \bar{X})]^2}{2\sigma^2}\right\} \cdot \frac{1}{\sigma^2} \\ &\propto \frac{1}{(\sigma^2)^{n/2+1}} \exp\left\{-\frac{\sum_{i=1}^n \{[Y_i - \hat{\beta}_1 - \hat{\beta}_2(X_i - \bar{X})] - (\beta_0 - \hat{\beta}_1) - (\beta_1 - \hat{\beta}_2)(X_i - \bar{X})\}^2}{2\sigma^2}\right\} \end{aligned}$$

REGRESSION

$$\begin{aligned}
 &\propto \frac{1}{(\sigma^2)^{n/2+1}} \exp\left\{-\frac{\sum_{i=1}^n [Y_i - \hat{\beta}_1 - \hat{\beta}_2(X_i - \bar{X})]^2 + \sum_{i=1}^n (\beta_0 - \hat{\beta}_1)^2}{2\sigma^2}\right\} \\
 &\cdot \exp\left\{-\frac{\sum_{i=1}^n (\beta_1 - \hat{\beta}_2)^2 (X_i - \bar{X})^2}{2\sigma^2}\right\} \cdot \\
 &\cdot \exp\left\{\frac{2 \sum_{i=1}^n [Y_i - \hat{\beta}_1 - \hat{\beta}_2(X_i - \bar{X})] (\beta_0 - \hat{\beta}_1)}{2\sigma^2}\right\} \cdot \\
 &\cdot \exp\left\{\frac{2 \sum_{i=1}^n [Y_i - \hat{\beta}_1 - \hat{\beta}_2(X_i - \bar{X})] (\beta_1 - \hat{\beta}_2)(X_i - \bar{X})}{2\sigma^2}\right\} \cdot \\
 &\cdot \exp\left\{-\frac{2 \sum_{i=1}^n (\beta_0 - \hat{\beta}_1)(\beta_1 - \hat{\beta}_2)(X_i - \bar{X})}{2\sigma^2}\right\}
 \end{aligned}$$

- All the double products will cancel, as we are going to see

REGRESSION

- $$\bullet \sum_{i=1}^n [Y_i - \hat{\beta}_1 - \hat{\beta}_2(X_i - \bar{X})] (\beta_0 - \hat{\beta}_1) = (\beta_0 - \hat{\beta}_1) \left\{ \sum_{i=1}^n [Y_i - \hat{\beta}_1] - \hat{\beta}_2 \sum_{i=1}^n (X_i - \bar{X}) \right\}$$

$$\Rightarrow (\beta_0 - \hat{\beta}_1) \left\{ \sum_{i=1}^n [Y_i - \bar{Y}] - \hat{\beta}_2 \sum_{i=1}^n (X_i - \bar{X}) \right\} = 0$$
- $$\bullet \sum_{i=1}^n [Y_i - \hat{\beta}_1 - \hat{\beta}_2(X_i - \bar{X})] (\beta_1 - \hat{\beta}_2)(X_i - \bar{X}) =$$

$$= (\beta_1 - \hat{\beta}_2) \left\{ \sum_{i=1}^n [Y_i - \hat{\beta}_1] (X_i - \bar{X}) - \hat{\beta}_2 \sum_{i=1}^n (X_i - \bar{X})^2 \right\} =$$

$$= (\beta_1 - \hat{\beta}_2) \left\{ \sum_{i=1}^n [Y_i - \bar{Y}] (X_i - \bar{X}) - \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} \sum_{i=1}^n (X_i - \bar{X})^2 \right\} = 0$$
- $$\bullet \sum_{i=1}^n (\beta_0 - \hat{\beta}_1)(\beta_1 - \hat{\beta}_2)(X_i - \bar{X}) = (\beta_0 - \hat{\beta}_1)(\beta_1 - \hat{\beta}_2) \sum_{i=1}^n (X_i - \bar{X}) = 0$$

REGRESSION

- Posterior $\pi(\beta_0, \beta_1, \sigma^2 | \underline{X}, \underline{Y})$

$$\propto \frac{1}{(\sigma^2)^{n/2+1}} \exp\left\{-\frac{\sum_{i=1}^n [Y_i - \hat{\beta}_1 - \hat{\beta}_2(X_i - \bar{X})]^2 + \sum_{i=1}^n (\beta_0 - \hat{\beta}_1)^2}{2\sigma^2}\right\} \cdot \exp\left\{-\frac{\sum_{i=1}^n (\beta_1 - \hat{\beta}_2)^2 (X_i - \bar{X})^2}{2\sigma^2}\right\}.$$

$$\propto \frac{1}{(\sigma^2)^{n/2+1}} \exp\left\{-\frac{SSR + n(\beta_0 - \hat{\beta}_1)^2 + (\beta_1 - \hat{\beta}_2)^2 \sum_{i=1}^n (X_i - \bar{X})^2}{2\sigma^2}\right\}$$

$$\propto \frac{1}{(\sigma^2)^{(n+1)/2}} \exp\left\{-\frac{SSR + n(\beta_0 - \hat{\beta}_1)^2}{2\sigma^2}\right\} \frac{1}{(\sigma^2)^{1/2}} \exp\left\{-\frac{(\beta_1 - \hat{\beta}_2)^2 \sum_{i=1}^n (X_i - \bar{X})^2}{2\sigma^2}\right\}.$$

- We integrate out $\beta_1 \sim \mathcal{N}(\hat{\beta}_2, \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2})$

$$\Rightarrow \pi(\beta_0, \sigma^2 | \underline{X}, \underline{Y}) \propto \frac{1}{(\sigma^2)^{(n+1)/2}} \exp\left\{-\frac{SSR + n(\beta_0 - \hat{\beta}_1)^2}{2\sigma^2}\right\}$$

REGRESSION

- We integrate out $\sigma^2 \sim \mathcal{IG}((n-1)/2, (SSR + n(\beta_0 - \hat{\beta}_1)^2)/2)$
- $\Rightarrow \beta_0 | \underline{X}, \underline{Y}$ will have a Student's t -distribution
- Student's t density $f(t)$ with mean 0, scale 1 and degrees of freedom (d.f.) ν

$$\Rightarrow f(t) \propto \frac{1}{\left[1 + \frac{t^2}{\nu}\right]^{(\nu+1)/2}}$$

$$\begin{aligned} \pi(\beta_0 | \underline{X}, \underline{Y}) &\propto \frac{1}{[SSR + n(\beta_0 - \hat{\beta}_1)^2]^{(n-1)/2}} \\ &\propto \frac{1}{\left[1 + \frac{(\beta_0 - \hat{\beta}_1)^2}{SSR/n}\right]^{[(n-2)+1]/2}} \\ &\propto \frac{1}{\left[1 + \frac{\frac{(\beta_0 - \hat{\beta}_1)^2}{S^2/n}}{n-2}\right]^{[(n-2)+1]/2}} \end{aligned}$$

REGRESSION

- I used $\frac{SSR}{n} = \frac{SSR}{n-2} \frac{n-2}{n} = S^2 \frac{n-2}{n}$ where S^2 is an unbiased estimator of σ^2
- $\Rightarrow \beta_0 | \underline{X}, \underline{Y} \sim t_{n-2}(\hat{\beta}_1, S^2/n)$, with mean $\hat{\beta}_1$, scale S^2/n and $(n-2)$ d.f.
- Similarly, it possible to prove that $\beta_1 | \underline{X}, \underline{Y} \sim t_{n-2}(\hat{\beta}_2, \frac{S^2}{\sum_{i=1}^n (X_i - \bar{X})^2})$
- We go back to $\pi(\beta_0, \sigma^2 | \underline{X}, \underline{Y}) \propto \frac{1}{(\sigma^2)^{n/2+1/2}} \exp\left\{-\frac{SSR + n(\beta_0 - \hat{\beta}_1)^2}{2\sigma^2}\right\}$
- We integrate out $\beta_0 \sim \mathcal{N}(\hat{\beta}_0, \sigma^2/n)$
- $\Rightarrow \pi(\sigma^2 | \underline{X}, \underline{Y}) \propto \frac{1}{(\sigma^2)^{n/2}} \exp\left\{-\frac{SSR}{2\sigma^2}\right\}$
- $\Rightarrow \sigma^2 | \underline{X}, \underline{Y} \sim \mathcal{IG}(n/2 - 1, SSR/2)$

REGRESSION

- Summarising, we have been able to get the three posteriors in closed form:
 - $\beta_0 | \underline{X}, \underline{Y} \sim t_{n-2}(\hat{\beta}_1, S^2/n)$
 - $\beta_1 | \underline{X}, \underline{Y} \sim t_{n-2}(\hat{\beta}_2, \frac{S^2}{\sum_{i=1}^n (X_i - \bar{X})^2})$
 - $\sigma^2 | \underline{X}, \underline{Y} \sim \mathcal{IG}(n/2 - 1, SSR/2)$
- Note that the posterior means for β_0 and β_1 coincide with the MLEs: this is not uncommon when considering improper priors
- Warning: the posteriors are proper if and only if $n > 2$ and not all X_i 's are equal!
- The posterior mean of σ^2 exists if and only if $n > 4$ since the mean of the inverse gamma $\mathcal{IG}(a, b)$ is $\frac{b}{a-1} = \frac{SSR/2}{(n/2-1)-1} = \frac{SSR}{n-4}$

REGRESSION

- We have found the posterior distributions of the parameters, either in closed form or in a suitable one to apply MCMC $\Rightarrow$ now we can estimate them, e.g., considering the posterior mean, and build credible intervals in a way similar to what we saw earlier (and I will not repeat it)
- When considering more than one covariate, i.e., $X_1, \dots, X_n$, still Gaussian priors should be considered for each of them
- Similarly to the frequentist approach, there is an interest for the covariates which are significant
 - Instead of considering p -values, Bayesians look for a credible interval and check if 0 belongs to it
 - If the credible interval contains 0 then the covariate is not significant; otherwise, it is
 - We will see an example next
- If Y is multivariate, then multivariate Gaussian distributions are chosen to model the observations and as a prior for the mean, while an Inverse Wishart distribution is chosen for the covariance matrix

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